On localization phenomena in plasticity models

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Part I

Necking of a metallic cylindrical bar

Chapter 1

Small deformation model

In this chapter we develop and analyze a small deformation model of a bar whose sections can only translate and change their size. In particular bending is not allowed. We split our discussion into three parts. The first section is devoted to the development of the model, the second to an existence and uniqueness results. The last section is devoted to the explicit solution of two boundary value problems describing the tensile testing of the bar. In both the cases we show that the small deformation model is not able to predict the necking phenomenon.

1.1 Formulation

1.1.1 Motivation from the multidimensional kinematics

In this section we describe the kinematics of the bar from a multidimensional perspective. This will serve as a motivation for the development of the purely one dimensional model discussed in the forthcoming sections.

Let $\Omega = (0, L) \times \Sigma$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ be the reference configuration of the cylindrical bar. We denote the generic point in Ω as (x, y) with $x \in (0, L)$ and $y \in \Sigma$. We assume a macroscopic displacement of the form

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} u_1(x) \\ u_2(x)y \end{pmatrix}$$

where u_1 and u_2 are functions from $(0, L)$ to R. This assumption means that the sections of the cylindrical bar remain flat and can translate and expand uniformly. The corresponding displacement gradient is

$$H = \begin{pmatrix} u_1' & 0 \\ u_2' y & u_2 I \end{pmatrix}.$$

The plastic distortion is assumed to be of the form

$$H_p = \begin{pmatrix} p & 0 \\ 0 & -\frac{p}{d-1} I \end{pmatrix}$$

where $p\colon (0, L) \to \mathbb{R}$. This choice implies plastic incompressibility. The elastic distortion is then given by

$$H_e = H - H_p = \begin{pmatrix} u_1' - p & 0 \\ u_2' y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix}.$$

Then the elastic strain is

$$E_e = \begin{pmatrix} u_1' - p & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix} = \begin{pmatrix} e_1 & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & e_2 I \end{pmatrix}$$

where we have introduced $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1} p$. Assuming an isotropic elastic response the elastic free energy density is given by

$$\psi_e = \frac{1}{2}\kappa(\operatorname{tr} E_e)^2 + \mu|\operatorname{dev} E_e|^2$$

where $\kappa \geqslant 0$ and $\mu \geqslant 0$ are the bulk and shear modulus respectively. We notice that

$$\operatorname{tr} E_e = e_1 + (d-1)e_2 \quad \text{and} \quad \operatorname{dev} E_e = \begin{pmatrix} \frac{d-1}{d}(e_1 - e_2) & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & -\frac{1}{d}(e_1 - e_2) I \end{pmatrix}.$$

Then the elastic free energy density can be written as

$$\psi_e = \frac{1}{2}\kappa\left(e_1 + (d-1)e_2\right)^2 + \frac{d-1}{d}\mu(e_1 - e_2)^2 + \frac{1}{2}|y|^2 \mu |u_2'|^2.$$

We conclude that the elastic free energy per unit axial length is

$$\int_\Sigma \psi_e dy = \frac{1}{2} A\kappa\left(e_1 + (d-1)e_2\right)^2 + \frac{d-1}{d} A\mu(e_1 - e_2)^2 + \frac{1}{2} J\mu |u_2'|^2$$

where

$$A = \int_\Sigma dy \quad \text{and} \quad J = \int_\Sigma |y|^2 dy$$

are the polar and the polar second order moment of area.

1.1.2 Kinematics

We now develop a purely one dimensional model of the bar taking inspiration from the multidimensional kinematics.

Let $L > 0$ be the length of the cylindrical bar. The macroscopic configuration of the bar is described by $u_i : (0, L) \to \mathbb{R}$ for $i \in \{1, 2\}$. Here u_1 is the longitudinal displacement of the sections in the axis direction and u_2 is the quantity describing the transversal expansion of the sections. The configuration of the microscopic structure of the bar is described by the plastic distortion $p : (0, L) \to \mathbb{R}$. We define $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1} p$ the elastic distortions.

1.1.3 Statics

We denote with $\tilde{v}$ and $\tilde{q}$ the virtual rates of u and p respectively. To describe the coupling between the macroscopic configuration u and the microscopic configuration p we choose an internal virtual power of the form

$$\tilde{W} = \int_0^L \zeta \tilde{v}_2' dx + \int_0^L \left(\sigma_1(\tilde{v}_1' - \tilde{q}) + \sigma_2(\tilde{v}_2 + (d-1)^{-1}\tilde{q})\right) dx + \int_0^L \tau \tilde{q} dx.$$

The first integral describes a purely macroscopic response.. Here ζ is the transversal stress. The second integral describes the elastic response that couples the macroscopic and the microscopic configurations. Here σ_1 is the longitudinal elastic stress and σ_2 the longitudinal elastic force. The last integral describes the microscopic response and τ is the microscopic force. The external longitudinal and transversal forces distributed along the length of the bar and the external longitudinal and transversal forces acting on the extremities of the bar are included choosing the external power

$$\tilde{\mathcal{P}} = \int_0^L f \cdot \tilde{v}\,dx + h(L) \cdot \tilde{v}(L) - h(0) \cdot \tilde{v}(0).$$

Proposition 1.1. The local form of the equilibrium condition is equivalent to

$$\sigma_1' + f_1 = 0 \quad \text{in } (0, L) \tag{1.1.1a}$$
$$h_1(x) = \sigma_1(x) \quad \text{for } x \in \{0, L\} \tag{1.1.1b}$$
$$-\sigma_2 + \zeta' + f_2 = 0 \quad \text{in } (0, L) \tag{1.1.1c}$$
$$h_2(x) = \zeta(x) \quad \text{for } x \in \{0, L\} \tag{1.1.1d}$$
$$\sigma_1 - (d-1)^{-1}\sigma_2 - \tau = 0 \quad \text{in } (0, L) \tag{1.1.1e}$$

Proof. First we choose $\tilde{v}_2 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_1$ arbitrary. Then we get

$$-\int_0^L \sigma_1'\tilde{v}_1\,dx + \sigma_1(L)\tilde{v}_1(L) - \sigma_1(0)\tilde{v}_1(0) = \int_0^L f_1\tilde{v}_1\,dx + h_1(L)\tilde{v}_1(L) - h_1(0)\tilde{v}_1(0).$$

We get (1.1.1a) and (1.1.1b). Next we choose $\tilde{v}_1 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_2$ arbitrary. Then

$$-\int_0^L \zeta'\tilde{v}_2\,dx + \zeta(L)\tilde{v}_2(L) - \zeta(0)\tilde{v}_2(0) + \int_0^L \sigma_2\tilde{v}_2\,dx = \int_0^L f_2\tilde{v}_2\,dx + h_2(L)\tilde{v}_2(L) - h_2(0)\tilde{v}_2(0).$$

We get (1.1.1c) and (1.1.1d). Then we choose $\tilde{v} = 0$ and $\tilde{q}$ arbitrary obtaining

$$\int_0^L (\sigma_1 - (d-1)^{-1}\sigma_2 - \tau)\tilde{q}\,dx = 0.$$

This implies (1.1.1e). □

1.1.4 Constitutive relations

We choose the free energy for unit axial length

$$\psi = \frac{1}{2}\kappa(e_1 + (d-1)e_2)^2 + \frac{1}{2}\mu(e_1 - e_2)^2 + \frac{1}{2}l^2\mu|u_2'|^2 + \frac{1}{2}Bp^2.$$

where κ is a bulk modulus, μ is a shear modulus, l is length parameter related to the geometry of the section and B is the hardening parameter. We also choose the dissipation potential for unit axial length

$$\phi = R(\dot{p})$$

where $R\colon \mathbb{R} \to [0, \infty]$ satisfy $R(0) = 0$. A possible choice could be $R(\dot{p}) = Y|\dot{p}|$ where $Y > 0$ is the yield strength. Consistently with these choices we impose the following constitutive relations

$$\begin{gathered}
\zeta = \mu l^2 u_2' \\
\sigma_1 = \kappa(e_1 + (d-1)e_2) + \mu(e_1 - e_2) \\
\sigma_2 = (d-1)\kappa(e_1 + (d-1)e_2) - \mu(e_1 - e_2) \\
\tau \in Bp + \partial R(\dot{p}).
\end{gathered}$$

Proposition 1.2. Every process that obeys the constitutive relations satisfy the dissipation inequality.

Proof. Let $\tau = Bp + \xi$ with $\xi \in \partial R(\dot{p})$. The dissipation per unit axial length is given by

$$\zeta \dot{u}_2' + \sigma_1 \dot{e}_1 + \sigma_2 \dot{e}_2 + \tau \dot{p} - \dot{\psi} = \xi \dot{p}.$$

By definition of subdifferential we have $0 = R(0) \geqslant R(\dot{p}) - \xi\dot{p}$ so that $\xi\dot{p} \geqslant R(\dot{p}) \geqslant 0$. This implies that the dissipation is nonnegative. □

We notice that introducing the constitutive relations into (1.1.1e) we get

$$\frac{d}{d-1}\mu(e_1 - e_2) - Bp \in \partial R(\dot{p}).$$

We refer to

$$m = \frac{d}{d-1}\mu(e_1 - e_2)$$

as the Mandel stress. It is the elastic force acting on the microscopic configuration and we notice that it depends on the shear modulus only.

1.2 Existence and uniqueness result

In this section we discuss the existence and uniqueness of solutions. For simplicity we will assume homogeneous Dirichlet boundary conditions $u(x) = 0$ for $x \in \{0, L\}$.

1.2.1 Function spaces and hypotheses

We define the spaces $\mathbf{U} = H^1((0,L), \mathbb{R} \times \mathbb{R})$ and $\mathbf{P} = L^2((0,L))$. Let also $\mathbf{Z} = \mathbf{U} \times \mathbf{P}$. We will write $z = (u, p) \in \mathbf{Z}$ with $u \in \mathbf{U}$ and $p \in \mathbf{P}$ and $w = (v, q) \in \mathbf{Z}$ with $v \in \mathbf{U}$ and $q \in \mathbf{P}$.

The existence and uniqueness result will be obtained under the following hypotheses.

(G) $L > 0$ and $d = 2$ or $d = 3$

(ψ) κ, μ, l and B are positive constants

(R) $R \colon \mathbb{R} \to \mathbb{R}$ is given by $R(q) = Y|q|$ for some $Y > 0$

(f) $f \in H^1((0,T), \mathbf{U}^*)$ and $f(0) = 0$

1.2.2 Formulation of the problem

We introduce the linear operator $\mathcal{A} \colon \mathbf{Z} \to \mathbf{Z}^*$ defined by

$$\begin{aligned}\langle \mathcal{A}z, w\rangle = \int_0^L l^2 \mu u_2' v_2' dx + \int_0^L \left[\kappa(u_1' + (d-1)u_2) + \mu\left(u_1' - u_2 - \frac{d}{d-1}p\right)\right](v_1' - q)dx + \\ + \int_0^L \left[(d-1)\kappa(u_1' + (d-1)u_2) - \mu\left(u_1' - u_2 - \frac{d}{d-1}q\right)\right]\left(v_2 + (d-1)^{-1}p\right) dx + \\ + \int_0^L Bpq\,dx\end{aligned}$$

and the functional $\mathcal{R} \colon \mathbf{Z} \to \mathbb{R} \cup \{\infty\}$ defined by

$$\mathcal{R}(w) = \int_0^L R(q)dx.$$

We define also $\mathcal{F}\colon [0,T] \to \mathbf{Z}^*$ by

$$\langle \mathcal{F}(t), w\rangle = \int_0^T f \cdot v \mathrm{d}x.$$

Now the boundary value problem associated to the model introduced in Section 1.1 can be formulated as follows.

Problem 1.3. Find $z = (u, p) \in AC([0,T], \mathbf{Z})$ such that $z(0) = 0$ and for almost every $t \in (0,T)$

$$\langle \mathcal{A}z(t), w - \dot{z}(t)\rangle + \mathcal{R}(w) - \mathcal{R}(\dot{z}(t)) \geqslant \langle \mathcal{F}(t), w - \dot{z}(t)\rangle \quad \text{for all } w \in \mathbf{Z}.$$

1.2.3 Existence and uniqueness theorem

To prove the existence and uniqueness result we make use of the following general theorem.

Theorem 1.4. Let $\mathbf{Z}$ be a reflexive and separable Banach space. Let $\mathcal{A}\colon \mathbf{Z} \to \mathbf{Z}^*$ be a linear and continuous functional which is also coercive, meaning that

$$\frac{\langle \mathcal{A}z, z\rangle}{\|z\|_{\mathbf{Z}}} \to \infty \quad \text{as } \|z\|_{\mathbf{Z}} \to \infty.$$

Let $\mathcal{R}\colon \mathbf{Z} \to \mathbb{R} \cup \{\infty\}$ be convex, proper, l.s.c., 1-homogeneous and Lipchitz continuous on its domain. Let $\mathcal{F} \in H^1((0,T), \mathbf{Z}^*)$ such that $\mathcal{F}(0) = 0$. Then there exists a unique $z \in AC([0,T], \mathbf{Z})$ such that for almost every $t \in (0,T)$

$$\langle \mathcal{A}z(t), w - \dot{z}(t)\rangle + \mathcal{R}(w) - \mathcal{R}(\dot{z}(t)) \geqslant \langle \mathcal{F}(t), w - \dot{z}(t)\rangle \quad \text{for all } w \in \mathbf{Z}.$$

Proof. See [22]. □

The main result of this section is the following theorem.

Theorem 1.5. Assume the hypotheses of Section 1.2.1. Then there exists a unique solution to Problem 1.3.

Proof. We verify the hypothesis of Theorem 1.4.

Step 1. It is trivial to check that $\mathcal{A}$ is linear and continuous. To prove coercivity we show that there exists $\alpha > 0$ such that $\langle \mathcal{A}z, z\rangle \geqslant \alpha\|z\|^2_{\mathbf{Z}}$. If we call $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1}p$ we have by applying Young inequality repeatedly

$$\begin{aligned}
\langle \mathcal{A}z, z\rangle = 2\int_0^L \psi \mathrm{d}x &= \kappa\|e_1 + (d-1)e_2\|_2^2 + \mu\|e_1 - e_2\|_2^2 + l^2\mu\|u_2'\|_2^2 + B\|p\|_2^2 = \\
&= \kappa\|u_1' + (d-1)u_2\|_2^2 + \mu\left\|u_1' - u_2 - \frac{d}{d-1}p\right\|_2^2 + l^2\mu\|u_2'\|_2^2 + B\|p\|_2^2 \geqslant \\
&\geqslant (1-\theta)\,\kappa\|u_1'\|_2^2 + \left(1 - \frac{1}{\theta}\right)(d-1)^2\kappa\|u_2\|_2^2 + (1-\tau)\mu\|u_1' - u_2\|_2^2 + \\
&\qquad l^2\mu\|u_2'\|_2^2 + \left[\left(1 - \frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2\mu + B\right]\|p\|_2^2 \\
&\geqslant [(1-\theta)\,\kappa + (1-\eta)(1-\tau)\mu]\,\|u_1'\|_2^2 + \\
&\quad + \left[\left(1 - \frac{1}{\theta}\right)(d-1)^2\kappa + \left(1 - \frac{1}{\eta}\right)(1-\tau)\mu\right]\|u_2\|_2^2 + l^2\mu\|u_2'\|_2^2 + \\
&\qquad + \left[\left(1 - \frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2\mu + B\right]\|p\|_2^2
\end{aligned}$$

for all θ, τ and η in $(0, \infty)$. We restrict to θ, τ and η in $(0, 1)$. Then by Poincarè inequality there exist $C_1 > 0$ and $C_2 > 0$ such that

$$\begin{aligned}\langle \mathcal{A}z, z\rangle \geqslant C_1 \left[(1-\theta)\,\kappa + (1-\eta)(1-\tau)\mu\right] \|u_1\|_{1,2}^2 + \\ + \left[C_2 l^2 \mu + \left(1 - \frac{1}{\theta}\right)(d-1)^2\kappa + \left(1 - \frac{1}{\eta}\right)(1-\tau)\mu\right] \|u_2\|_{1,2}^2 + \\ + \left[\left(1 - \frac{1}{\tau}\right)\left(\frac{d}{d-1}\right)^2 \mu + B\right] \|p\|_2^2\end{aligned}$$

Now we choose θ, τ and η sufficiently close to 1 so that the coefficients multiplying $\|u_1\|_{1,2}^2$, $\|u_2\|_{1,2}^2$ and $\|p\|_2^2$ are positive and we get the desired coercivity.

Step 2. Since $\mathcal{R}(w) = Y\|q\|_1$ we see immediately that $\mathcal{R}$ is convex, proper, continuous and 1-homogeneous.

Step 3. Since $f \in H^1((0, T), \mathbf{U}^*)$ and $f(0) = 0$ then $\mathcal{F} \in H^1((0, T), \mathbf{Z}^*)$ and $\mathcal{F}(0) = 0$. □

1.3 Some analytical solutions

1.3.1 Traction with free end sections

Suppose that $f(x, t) = 0$, $h_1(L, t) = h_1(0, t) = F(t)$, and $h_2(L, t) = h_2(0, t) = 0$ for all $x \in (0, L)$ and $t \in [0, T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. From equation (1.1.1a) and (1.1.1b) we have that $\sigma_1 = F$ on all $(0, L)$. Then

$$\kappa(e_1 + (d-1)e_2) = F - \mu(e_1 - e_2) \quad \text{and} \quad e_1 = \frac{F - (d-1)\kappa e_2 + \mu e_2}{\kappa + \mu}.$$

Moreover

$$e_1 - e_2 = \frac{F - d\kappa e_2}{\kappa + \mu}.$$

Then

$$\begin{aligned}\sigma_2 &= (d-1)\left(F - \mu(e_1 - e_2)\right) - \mu(e_1 - e_2) = (d-1)F - d\mu(e_1 - e_2) = \\ &= (d-1)F - d\frac{\mu}{\kappa+\mu}F + d^2\frac{\kappa\mu}{\kappa+\mu}e_2 = \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + d^2\frac{\kappa\mu}{\kappa+\mu}e_2 = \\ &= \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}p + d^2\frac{\kappa\mu}{\kappa+\mu}u_2.\end{aligned}$$

Now from equation (1.1.1c) we have

$$l^2\mu u_2'' = \frac{(d-1)\kappa - \mu}{\kappa+\mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}p + d^2\frac{\kappa\mu}{\kappa+\mu}u_2.$$

The solution is given by

$$u_2 = -\frac{p}{d-1} - \frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

since conditions $h_2(L, t) = h_2(0, t)$ are satisfied because $u_2' = 0$. Notice that

$$e_2 = u_2 + \frac{p}{d-1} = -\frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

Now the Mandel stress is given by

$$m = \frac{d}{d-1}\frac{\mu}{\kappa+\mu}(F - d\kappa e_2) = F.$$

When $F \leqslant Y$ then the response is purely elastic and $p = 0$. When $F > Y$ then $\dot{p} > 0$ and

$$F - Bp = Y.$$

This implies $p = B^{-1}(F - Y)$. Assuming $(d-1)\kappa - \mu > 0$ the sections constricts in traction.

1.3.2 Traction with clamped end sections

Setup of the problem

Suppose that $f(x,t) = 0$, $h_1(L,t) = h_1(0,t) = F(t)$, and $u_2(L,t) = u_2(0,t) = 0$ for all $x \in (0,L)$ and $t \in [0,T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. By the same computations of Section 1.3.1 we find the equation

$$l^2\mu u_2'' = \frac{(d-1)\kappa-\mu}{\kappa+\mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}p + d^2\frac{\kappa\mu}{\kappa+\mu}u_2. \tag{1.3.1}$$

Solution in the elastic range

In the elastic range $p = 0$ everywhere in $(0,L)$. In this case a particular solution is given by

$$x \mapsto -\frac{(d-1)\kappa-\mu}{d^2\kappa\mu}F$$

and a solution of the homogeneous equation which is also symmetric with respect to $x = L/2$ is

$$x \mapsto \cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right).$$

Then taking into account the boundary conditions we find the solution

$$u_2 = e_2 = \frac{(d-1)\kappa-\mu}{d^2\kappa\mu}\left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)} - 1\right)F. \tag{1.3.2}$$

When $(d-1)\kappa - \mu > 0$ the sections constricts in traction. The middle section is the most affected by the constriction. The Mandel stress is the elastic range is given by

$$m = \frac{d}{d-1}\frac{\mu}{\kappa+\mu}(F - d\kappa e_2) = \frac{d}{d-1}\frac{\mu}{\kappa+\mu}\left(1 - \frac{(d-1)\kappa-\mu}{d\mu}\left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)} - 1\right)\right)F.$$

Its maximum is reached in the middle section so we conclude that the Mandel stress reach yield strength here first.

Solution in the plastic range

From the above considerations we are let to seek for a solution where $p > 0$ in the interval $(L/2 - \epsilon, L/2 + \epsilon)$ where $\epsilon \in [0, L/2]$ is a function of time.

Remark 1.6. Since the solution is symmetric with respect to $x = L/2$ from now on we focus on the $(0, L/2)$ half of the bar only.

On $(0, L/2 - \epsilon)$ the solution is given by $p = 0$ and u_2 of the form

$$u_2 = \frac{(d-1)\kappa - \mu}{d^2\kappa\mu}\left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{x-\frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)} - 1\right)F + C_1 \sinh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}x\right) \tag{1.3.3}$$

where $C_1 \in \mathbb{R}$. On $(L/2 - \epsilon, L/2)$ we have

$$m - Bp = Y$$

$$\frac{d}{d-1}\frac{\mu}{\kappa+\mu}(F - d\kappa u_2 - \frac{d\kappa}{d-1}p) - Bp = Y.$$

Then we find

$$p = \frac{\frac{d}{d-1}\frac{\mu}{\kappa+\mu}F - \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa+\mu}u_2 - Y}{\frac{d^2}{(d-1)^2}\frac{\mu\kappa}{\kappa+\mu} + B}.$$

Substituting that identity into equation (1.3.1) we have obtain a differential equation in y of the form

$$y'' = ay + b$$

where a and b depends on time only. A solution which is symmetric with respect to $x = L/2$ is of the form

$$y = -\frac{b}{a} + C_2 \cosh\left(\sqrt{a}\left(x - \frac{L}{2}\right)\right). \tag{1.3.4}$$

Remark 1.7. Since the computations became to long to be done by hand we perform them with an automatic algebraic manipulator (the MATLAB Symbolic Math Toolbox). Since most of the resultant expressions are too long we do not report them.

We now look for a solution $u_2 \in C^2([0, L])$. When $\epsilon = L/2$ the solution is simply given by

$$u_2 = \frac{b}{a}\left(\frac{\cosh\left(\sqrt{a}\left(x - \frac{L}{2}\right)\right)}{\cosh\left(\sqrt{a}\frac{L}{2}\right)} - 1\right)$$

When $\epsilon \in (0, L/2)$ we consider (1.3.3) on $(0, L/2 - \epsilon)$ and (1.3.4) on $(L/2 - \epsilon, L/2)$ and we impose the continuity of u_2 and u_2' at $x = L/2 - \epsilon$. Solving the resultant linear system we find the constants C_1 and C_2.

Determination of ϵ

We are left with the determination of $\epsilon \in [0, L/2]$ as a function of time. To this aim we consider the Mandel stress computed the elastic range at $x = L/2$. We call F_Y the value of F for which that Mandel stress reach Y. We find

$$F_Y = \frac{d-1}{d}\frac{k+\mu}{\mu}\left(-\frac{(d-1)\kappa - \mu}{d\mu}\left(\frac{1}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}}\frac{L}{2l}\right)} - 1\right) + 1\right)^{-1} Y.$$

For $F \leqslant F_Y$ we have $\epsilon = 0$. To determine ϵ when $F > F_Y$ we consider the equation $m(L/2 - \epsilon) = Y$. That equation cannot be solved analytically in ϵ. Instead we can find explicitly F as a function $f =: [0, \epsilon_0) \to \mathbb{R}$ of ϵ. Here $[0, \epsilon_0)$ is the maximal interval where f is well defined. After a meticulous analysis of the analytical expression of f we find that $\epsilon_0 > L/2$. This means that when the load F reach the value

$$f(L/2) = \frac{d-1}{d}\frac{\kappa+\mu}{\mu}Y,$$

the plastic distortion satisfies $p > 0$ in the entire interval $(0, L)$. We conclude that

$$\epsilon = \begin{cases} 0 & \text{if } F \leqslant F_Y \\ f^{-1}(F) & \text{if } F_Y < F \leqslant f(L/2) \\ L/2 & \text{if } F > f(L/2). \end{cases}$$

Absence of necking

From the analytical expression one can see that for $F > f(L/2)$ the solution u_2 depends linearly on F. In particular as $F \to \infty$ we have that $u_2 \to -\infty$ uniformly on all compact sets contained in $(0, L)$. We conclude that the small deformation model is not able to predict necking. Indeed the shrinkage of the cross section seems to be not localized on a small region around the middle of the bar but to be distributed along its entire length.

The limit of thin bar

To get further insight on the solution we look at the limit $l \to 0^+$.

Proposition 1.8. In the limit $l \to 0^+$ the solution u_2 that we have found converges pointwise on $(0, L) \times [0, T]$ to the homogeneous solution of Section 1.3.1. This convergence is also uniform on compact subsets of $(0, L) \times [0, T]$.

Proof. It is a direct computation. □

This results confirm the absence of necking in the small deformation model. Indeed it says that a thin bar deforms almost uniformly.

Some plots

For completeness we report some plots illustrating the solution that we have found. We fix $d = 3$, $L = 1$, $l = 0.1$, $\kappa = 100$, $\mu = 100$, $B = 0.1$, $Y = 1$. In Figure 1.1 on the left we port the graph of f. We see that ϵ_0, which correspond to the abscissa of the vertical asymptote, is slightly greater then $L/2$. In Figure 1.2 we report the plot of u_2 and p when $\epsilon = 0.3$. In Figure 1.3 we report a picture of the bar when $\epsilon = 0.493$. On the top there is the result obtained with a purely elastic bar while on the bottom the result predicted including the plastic behavior. The displacements of cross sections is magnified with a factor 7. The color represents the variable p.

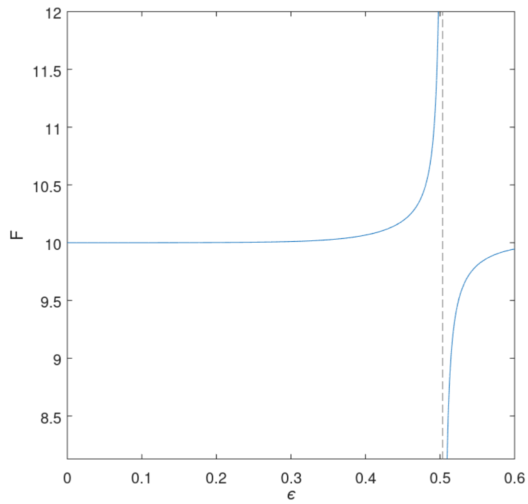

Figure 1.1: Graph of the function f.

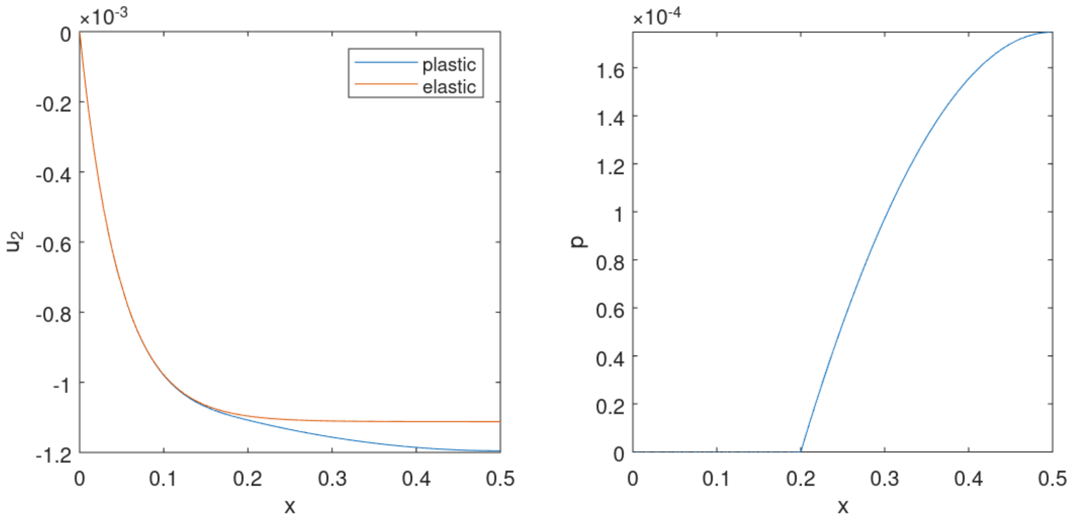

Figure 1.2: Plot of u_2 (left) and p (right) for $\epsilon = 0.3$.

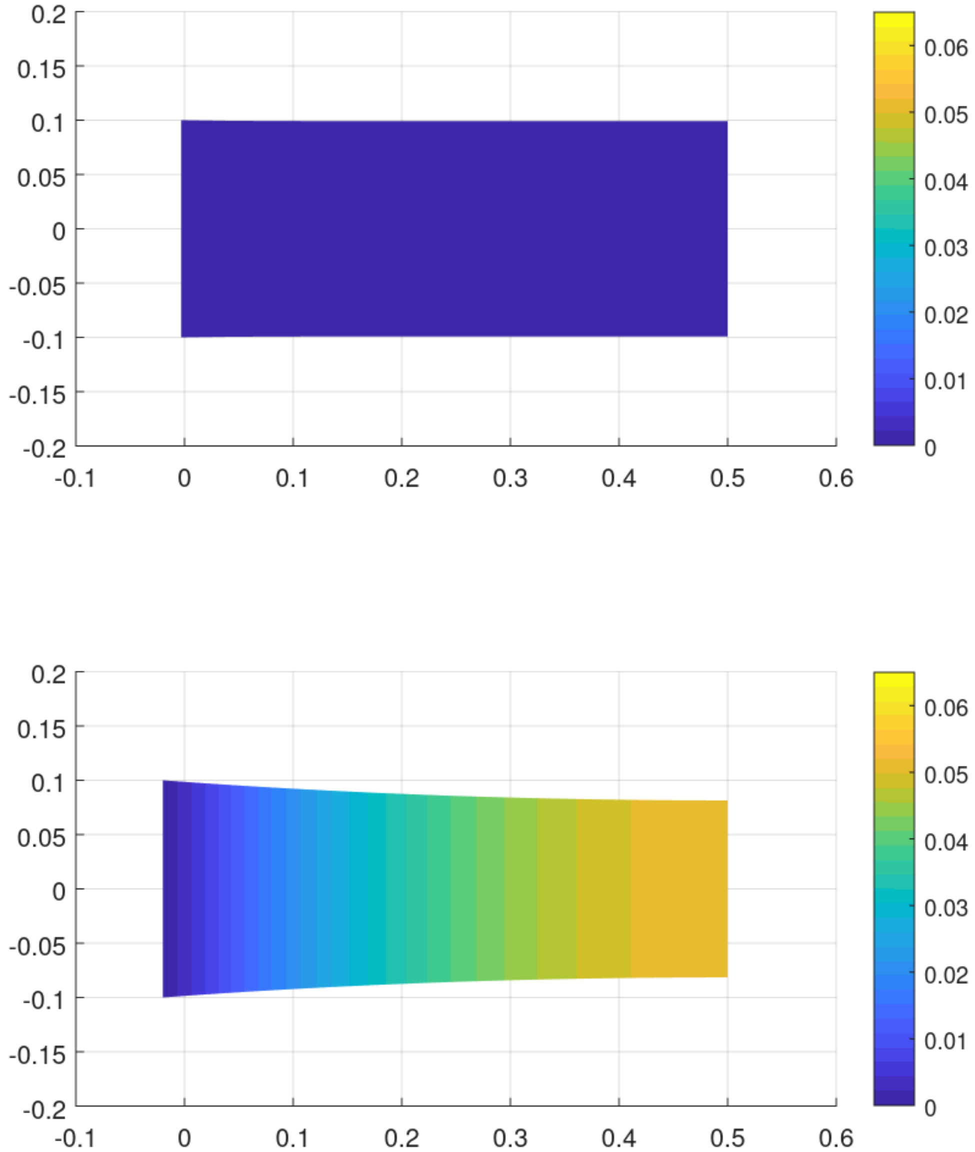

Figure 1.3: Picture of the half $(0, L/2)$ of the bar when $\epsilon = 0.493$. The bar on the bottom is plastic, the one on the top is purely elastic and subject to the same load. The transversal displacement is magnified with a factor 7. The color represents p.

Chapter 2

Large deformation model

As we showed in the previous chapter, a small strain model of a bar is not able to predict necking. This leads us to consider a geometrically exact model that allows large deformations.

2.1 Formulation

2.1.1 Motivation form the multidimensional kinematics

Let $\Omega = (0, L) \times \Sigma \subset \mathbb{R}^d$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ the reference configuration of the cylindrical bar. The generic material point will be denoted as $(X, Y) \in \Omega$ with $X \in (0, L)$ and $Y \in \Sigma$. We consider deformations of the form

$$(X, Y) \mapsto \begin{pmatrix} \chi_1(X) \\ \chi_2(X)Y \end{pmatrix}$$

where $\chi_1 \colon (0, L) \to \mathbb{R}$ satisfy $\chi_1' > 0$ and $\chi_2 \colon (0, L) \to (0, \infty)$. The corresponding deformation gradient is

$$F = \begin{pmatrix} \chi_1' & 0 \\ \chi_2' Y & \chi_2 I \end{pmatrix}.$$

We define $\epsilon_1 = \log \chi_1'$ and $\epsilon_2 = \log \chi_2$. The plastic distortion is assumed to be of the form

$$F_p = \begin{pmatrix} P & 0 \\ 0 & P^{-\frac{1}{d-1}} I \end{pmatrix}$$

where $P \colon (0, L) \to (0, \infty)$. This choice implies plastic incompressibility. We define $p = \log P$. The elastic distortion is then given by

$$F_e = FF_p^{-1} = \begin{pmatrix} \chi_1' P^{-1} & 0 \\ \chi_2' P^{-1} Y & \chi_2 P^{\frac{1}{d-1}} I \end{pmatrix} = \begin{pmatrix} E_1 & 0 \\ GY & E_2 I \end{pmatrix}.$$

We define

$$e_1 = \log E_1 = \epsilon_1 - p \quad \text{and} \quad e_2 = \log E_2 = \epsilon_2 + (d-1)^{-1} p.$$

Now the spatial velocity gradient is given by

$$L = \dot{F} F^{-1} = \begin{pmatrix} \dot{\epsilon}_1 & 0 \\ \left(\frac{\dot{\chi}_2'}{\chi_1'} - \frac{\chi_2'}{\chi_1'} \dot{\epsilon}_2 \right) & \dot{\epsilon}_2 I \end{pmatrix}$$

and the plastic rate is

$$L_p = \dot{F}_p F_p^{-1} = \begin{pmatrix} \dot{p} & 0 \\ 0 & -\frac{\dot{p}}{d-1} I \end{pmatrix}.$$

Then the elastic rate is

$$L_e = \dot{F}_e F_e^{-1} = L - F_e L_p F_e^{-1} = \begin{pmatrix} \dot{e}_1 & 0 \\ \left(\frac{\dot{G}}{E_1} - \frac{G}{E_1}\dot{e}_2\right) Y & \dot{e}_2 I \end{pmatrix}.$$

Consider the case $d = 2$. In this case the first principal invariant of $C_e = F_e^T F_e$ is

$$I_e = \mathrm{tr}(C_e) = E_1^2 + G^2|Y|^2 + E_2^2$$

and

$$J_e = \det F_e = E_1 E_2.$$

The compressible neo-Hookean constitutive model is based on the elastic free energy density

$$\psi_e = C\left(\frac{I_e}{J_e} - 2\right) + w(J) = C\left(\frac{E_1^2 + G^2|Y|^2 + E_2^2}{E_1 E_2} - 2\right) + w(E_1 E_2)$$

2.1.2 Kinematics

Let $L > 0$ the length of the cylindrical bar. The macroscopic configuration of the bar is described by $\chi_1 \colon (0, L) \to \mathbb{R}$ with $\chi_1' > 0$ and $\chi_2 \colon (0, L) \to (0, \infty)$. Here χ_1 describes the positions of the sections and χ_2 is the transversal deformation factor of the sections. The configuration of the microscopic structure of the bar, that is the plastic distortion, is described by $P \colon (0, L) \to (0, \infty)$. We define the elastic distortions as $E_1 = \chi_1' P^{-1}$, $E_2 = \chi_2 P^{\frac{1}{d-1}}$ and $G = \chi_2' P^{-1}$.

2.1.3 Statics

We assume that the internal power is given by

$$\mathcal{W} = \int_0^L \left(\zeta\dot{G} + \sigma_1\dot{E}_1 + \sigma_2\dot{E}_2\right) dX + \int_0^L \left(\tau\dot{P} + \kappa\dot{P}'\right) dX.$$

Here ζ is the transversal elastic stress, σ_1 is the longitudinal elastic stress and σ_2 is the transversal elastic force while τ is the microscopic force and κ the microscopic stress.

Remark 2.1. In the small strain model of Chapter 1 we did not included the microscopic stress κ. However, we are now including it because we expect that in the large-strain regime, it is necessary to obtain an existence result in Sobolev spaces. This is suggested by the existence theorem for three-dimensional elastoplasticity at large strains presented in [23].

We notice that

$$\begin{aligned} G^{-1}\dot{G} &= (\chi_2')^{-1}\dot{\chi}_2' - P^{-1}\dot{P} \\ E_1^{-1}\dot{E}_1 &= (\chi_1')^{-1}\dot{\chi}_1' - P^{-1}\dot{P} \\ E_2^{-1}\dot{E}_2 &= \chi_2^{-1}\dot{\chi}_2 + (d-1)^{-1}P^{-1}\dot{P}. \end{aligned}$$

Then if we call $\tilde{v}_i$ the virtual rate of χ_i and $\tilde{Q}$ the virtual rate of P we are let to consider the internal virtual power

$$\begin{aligned}\tilde{\mathcal{W}} = \int_0^L G\zeta\left((\chi_2')^{-1}\tilde{v}_2' - P^{-1}\tilde{Q}\right) dX + \\ + \int_0^L \left[E_1\sigma_1\left((\chi_1')^{-1}\tilde{v}_1' - P^{-1}\tilde{Q}\right) + E_2\sigma_2\left(\chi_2^{-1}\tilde{v}_2 + (d-1)^{-1}P^{-1}\tilde{Q}\right)\right] dX + \\ + \int_0^L \left[\tau\tilde{Q} + \kappa\tilde{Q}'\right] dX.\end{aligned}$$

The external longitudinal and transversal forces distributed along the length of the bar and the external longitudinal and transversal forces acting on the extremities of the bar are included choosing the external power

$$\tilde{\mathcal{P}} = \int_0^L f \cdot \tilde{v} dx + h(L) \cdot \tilde{v}(L) - h(0) \cdot \tilde{v}(0).$$

Proposition 2.2. The local form of the equilibrium condition is

$$\begin{gathered}(P^{-1}\sigma_1)' + f_1 = 0 \quad \text{in } (0, L) \\ h_1(x) = P(x)^{-1}\sigma_1(x) \quad \text{for } x \in \{0, L\} \\ -P^{\frac{1}{d-1}}\sigma_2 + (P^{-1}\zeta)' + f_2 = 0 \quad \text{in } (0, L) \\ h_2(x) = P(x)^{-1}\zeta(x) \quad \text{for } x \in \{0, L\} \\ \chi_1' P^{-2}\sigma_1 - (d-1)^{-1}\chi_2 P^{\frac{1}{d-1}-1}\sigma_2 + \chi_2' P^{-2}\zeta - \tau + \kappa' = 0 \quad \text{for } x \in \{0, L\} \\ \kappa(x) = 0 \quad \text{for } x \in \{0, L\}\end{gathered}$$

Proof. Rearranging the terms we have

$$\begin{aligned}\tilde{\mathcal{W}} = \int_0^L \left[P^{-1}\sigma_1\tilde{v}_1' + P^{\frac{1}{d-1}}\sigma_2\tilde{v}_2 + P^{-1}\zeta\tilde{v}_2'\right] dX + \\ + \int_0^L \left[\left(\tau - \chi_1' P^{-2}\sigma_1 + (d-1)^{-1}\chi_2 P^{\frac{1}{d-1}-1}\sigma_2 - \chi_2' P^{-2}\zeta\right)\tilde{Q} + \kappa\tilde{Q}'\right] dX.\end{aligned}$$

By integration by parts and application of the fundamental lemma of the calculus of variation we get the thesis. □

2.1.4 Constitutive relations

We choose a free energy for unit axial length of the form

$$\psi = W(E_1, E_2, G) + H(P, P')$$

where $W : (0, \infty) \times (0, \infty) \times \mathbb{R} \to \mathbb{R}$ is elastic free energy density and $H : (0, \infty) \times \mathbb{R} \to \mathbb{R}$ is the hardening free energy density. The plastic response is obtained by introducing a dissipation potential per unit axial length of the form

$$\phi = R(P^{-1}\dot{P})$$

where $R : \mathbb{R} \to [0, \infty]$ satisfies $R(0) = 0$. Consistently with that choice of ψ and ϕ we impose the following constitutive relations

$$\sigma_i = \frac{\partial W}{\partial E_i} \quad \text{and} \quad \zeta = \frac{\partial W}{\partial G} \tag{2.1.1a}$$

$$\tau \in \frac{\partial H}{\partial P} + P^{-1}\partial R(P^{-1}\dot{P}) \quad \text{and} \quad \kappa = \frac{\partial H}{\partial P'}. \tag{2.1.1b}$$

Proposition 2.3. The constitutive relations (2.1.1) satisfy the dissipation inequality.

Proof. Let $\tau = \frac{\partial H}{\partial P} + P^{-1}\xi$ with $\xi \in \partial R(\dot{P}P^{-1})$. Then dissipation per unit axial length is given by

$$\zeta\dot{G} + \sigma_2\dot{E}_2 + \sigma_2\dot{E}_2 + \tau\dot{P} + \kappa\dot{P}' - \dot{\psi} = \xi P^{-1}\dot{P}.$$

By definition of subdifferential we have $0 = R(0) \geqslant R(P^{-1}\dot{P}) - \xi P^{-1}\dot{P}$ so that $\xi P^{-1}\dot{P} \geqslant R(P^{-1}\dot{P}) \geqslant 0$. This implies that the dissipation is nonnegative. □

Particular constitutive relations, can be obtained taking inspiration from the neo-Hookeian constitutive model. Specifically we choose the elastic free energy density

$$W(E_1, E_2, G) = C\left(\frac{E_1^2 + l^2G^2 + E_2^2}{E_1E_2} - 2\right) + w(E_1E_2).$$

In this case

$$\sigma_1 = C\left(\frac{1}{E_2} - \frac{l^2G^2}{E_1^2E_2} - \frac{E_2}{E_1^2}\right) + E_2w'(E_1E_2), \quad \sigma_2 = C\left(-\frac{E_1}{E_2^2} - \frac{l^2G^2}{E_1E_2^2} + \frac{1}{E_1}\right) + E_1w'(E_1E_2)$$

$$\zeta = 2C\frac{l^2G}{E_1E_2}$$

2.2 Tensile test

Equations in the elastic range

$$C\left(\frac{1}{\chi_2} - \frac{l^2(\chi_2')^2}{(\chi_1')^2\chi_2} - \frac{\chi_2}{(\chi_1')^2}\right) + \chi_2w'(\chi_1'\chi_2) = F$$

$$-C\left(-\frac{\chi_1'}{\chi_2^2} - \frac{l^2(\chi_2')^2}{\chi_1'\chi_2^2} + \frac{1}{\chi_1'}\right) - \chi_1'w'(\chi_1'\chi_2) + 2Cl^2\left(\frac{\chi_2'}{\chi_1'\chi_2}\right)' = 0.$$

From the first equation we get the following third order algebraic equation for χ_1'

$$2A\chi_2^2(\chi_1')^3 + \left(\frac{C}{\chi_2} - 2A\chi_2 - F\right)(\chi_1')^2 - C\left(l^2\frac{(\chi_2')^2}{\chi_2} + \chi_2\right) = 0.$$

Part II

Shear band localization in cohesive materials

Chapter 3

Small deformation model

3.1 Formulation

In this section we formulate the mechanical model. The description is made very quickly and we will expand it to give a better justification. In this section the emphasis is on the mechanical modeling, the correct functional setting will be investigated in the subsequent sections.

3.1.1 Kinematics

Let $\Omega \subset \mathbb{R}^d$ be the reference configuration of the body. The macroscopic displacement is a map $u\colon \Omega \to \mathbb{R}^d$. We set $E = \operatorname{sym}\operatorname{grad} u$ and $M = \operatorname{grad}^2 u$. We subdivide $\partial\Omega$ in two disjoint parts Γ_D and Γ_N. On Γ_D we impose the kinematical constrain $u = \overline{u}$. The configuration of the microscopic structure is described by the plastic distortion $H_p\colon \Omega \to \mathrm{Lin}$. The skew symmetric part $W_p = \operatorname{skew} H_p$ describes the rotation of the microscopic structure and for simplicity we assume that it vanish. The symmetric part is the plastic strain $E_p = \operatorname{sym} H_p\colon \Omega \to \mathrm{Sym}$ which describes the deformation of microscopic structure. We can regard E_p as the elastically relaxed strain of the body so we define the elastic strain as $E_e = E - E_p$. The volumetric deformation of the microscopic structure is described by $\lambda_p = \operatorname{tr} E_p$ while the deviatoric deformation by $U_p = \operatorname{dev} E_p$. We call $M_p = \operatorname{grad} U_p$ and we set $\mathbb{U}_p = (U_p, l\,M_p) \in \mathrm{DevPla}$ where $l > 0$ is an internal length scale. Here $\mathrm{DevPla} = \mathrm{Sym} \otimes (\mathrm{Sym} \otimes \mathbb{R})$ is the vector space of the deviatoric plastic strains and their gradients. In what follows we will denote with $\circ$ the natural scalar product on DevPla. We assume that the material is not dilatant (this is a simplification and should be generalized to describe the dilatant behavior observed in most materials exhibiting a pressure dependent yield strength) so we add the constrain $\lambda_p = 0$. To impose this constrain we add the Lagrange multiplier $p\colon \Omega \to \mathbb{R}$ with the meaning of a plastic pressure. The constrain can be written in weak form as

$$\int_\Omega \lambda_p \tilde{q}\, dx = 0 \quad \text{for all } \tilde{q}\colon \Omega \to \mathbb{R}.$$

3.1.2 Statics

We denote with $\tilde{v}$, $\tilde{L}_p$ $\tilde{V}_p$ and $\tilde{\mu}_p$ the virtual rates of u, E_p, U_p and λ_p respectively. The virtual rate of E_e is $\tilde{L}_e = \operatorname{grad}\tilde{v} - \tilde{L}_p$ where $\tilde{L} = \operatorname{grad}\tilde{v}$. Due to the presence of the Dirichlet condition we require that $\tilde{v}$ vanish on Γ_D. Instead we do not require that $\tilde{\mu}_p = 0$ because the constrain $\lambda_p = 0$ is imposed by mean of the Lagrange multiplier p. To describe the coupling

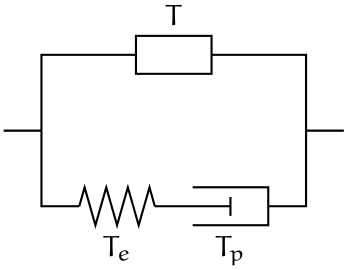

Figure 3.1: Rheological model describing the choice of the internal virtual power $\tilde{\mathcal{W}}$. The box in upper branch represents the (generic) macroscopic response. In the lower branch, the spring represents the elastic response and the piston represents the microscopic response. We notice that the microscopic response besides the usual dissipative plastic microforce force can include conservative microforces to model hardening. A more detailed discussion on constitutive relations is in Section 3.1.3.

between the macroscopic deformation u and the microscopic configuration E_p we choose an internal virtual power of the form

$$\tilde{\mathcal{W}} = \int_\Omega \left(T : \tilde{L} + K \,\vdots\, \mathrm{grad}^2\, \tilde{v} \right) dx + \int_\Omega T_e : \tilde{L}_e \, dx + \int_\Omega \left(T_p : \tilde{V}_p + K_p \,\vdots\, \mathrm{grad}\, \tilde{V}_p \right) dx.$$

The first integral is related to the macroscopic response. Here $T \in \mathrm{Sym}$ is the macrostress and $K \in \mathrm{Lin} \otimes \mathbb{R}$ is the macrohyperstress. The presence of the second order of $\tilde{v}$ is due to the need of a sufficient regularization in view of the existence theorem. The second integral describe the coupling between the macroscopic and the microscopic response and $T_e \in \mathrm{Sym}$ is the elastic stress. The third integral is associated with the microscopic response. Here $T_p \in \mathrm{DevSym}$ is the microforce and $K_p \in \mathrm{DevSym} \otimes \mathbb{R}$ is the microstress. In Figure 3.1 we report a picture that represents schematically the meaning of the internal virtual power that we have chosen. We define $\tilde{\mathbb{V}}_p = (\tilde{V}_p, l\, \mathrm{grad}\, \tilde{V}_p)$ and $\mathbb{T}_p = (T_p, l^{-1} K_p)$ so that

$$T_p : \tilde{V}_p + K_p \,\vdots\, \mathrm{grad}\, \tilde{V}_p = \mathbb{T}_p \circ \tilde{\mathbb{V}}_p.$$

The action of the plastic pressure is obtained by considering the reaction virtual power

$$\tilde{\mathcal{R}} = -\int_\Omega p \tilde{\mu}_p \, dx.$$

We notice that this power vanish on virtual rates that satisfy the constrain that is $\tilde{\mu}_p = 0$. The external actions of the volume force f on Ω and the traction h on Γ_N are included choosing the external virtual power

$$\tilde{\mathcal{P}} = \int_\Omega f \cdot \tilde{v} dx + \int_{\Gamma_N} h \cdot \tilde{v} dx.$$

For simplicity we assume that no external torque is applied to the body and that there is no external action on the microscopic configuration. These external actions could be added to describe more accurately the boundary conditions imposed on the body.

Lemma 3.1. The local form of the equilibrium condition is

$$
\begin{aligned}
&\operatorname{div}(T + T_e - \operatorname{div} K) + f = 0 \quad \text{in } \Omega\\
&(T + T_e - \operatorname{div} K)n - \operatorname{div}_P((Kn)P) = h \quad \text{in } \Gamma_N \cap \partial^{d-1}\Omega\\
&\{\!\{(Kn)\tau\}\!\} = 0 \quad \text{in } \partial^{d-2}\Omega\\
&-\operatorname{dev} T_e + T_p - \operatorname{div} K_p = 0 \quad \text{in } \Omega\\
&p = -\frac{1}{d} \operatorname{tr} T_e \quad \text{in } \Omega\\
&K_p n = 0 \quad \text{in } \partial^{d-1}\Omega.
\end{aligned}
$$

Proof. By integration by parts we can write

$$
\begin{aligned}
\int_\Omega &\left(T : \operatorname{grad} \tilde{v} + K \,\vdots\, \operatorname{grad}^2 \tilde{v}\right) dx = \\
&= -\int_\Omega \operatorname{div}(T - \operatorname{div} K) \cdot \tilde{v} dx + \int_{\partial^{d-1}\Omega} \left[(T - \operatorname{div} K)n \cdot \tilde{v} + Kn : \operatorname{grad} \tilde{v}\right] dx.
\end{aligned}
$$

Now denoting with $P = I - n \otimes n$ the projector on $\partial^{d-1}\Omega$ we have

$$
Kn : \operatorname{grad} \tilde{v} = Kn : \left(\frac{\partial \tilde{v}}{\partial n} \otimes n + \operatorname{grad}_P \tilde{v}\right) = (Kn)n \cdot \frac{\partial \tilde{v}}{\partial n} + Kn : \operatorname{grad}_P \tilde{v}
$$

and

$$
Kn : \operatorname{grad}_P \tilde{v} = (Kn)P : \operatorname{grad}_P \tilde{v} = \operatorname{div}_P(P(Kn)^T \tilde{v}) - \operatorname{div}_P((Kn)P) \cdot \tilde{v}.
$$

But by integration by parts

$$
\int_{\partial^{d-1}\Omega} \operatorname{div}_P(P(Kn)^T \tilde{v}) dx = \int_{\partial^{d-2}\Omega} \{\!\{P(Kn)^T \tilde{v} \cdot \tau\}\!\} dx = \int_{\partial^{d-2}\Omega} \{\!\{(Kn)\tau\}\!\} \cdot \tilde{v} dx.
$$

Here $\{\!\{\cdot\}\!\}$ denotes sum of its argument evaluated on both sides of $\partial^{d-2}\Omega$ and τ denotes the unit normal vector to $\partial^{d-2}\Omega$ and outer tangent to $\partial^{d-1}\Omega$. We have obtained

$$
\begin{aligned}
\int_\Omega &\left(T : \operatorname{grad} \tilde{v} + K \,\vdots\, \operatorname{grad}^2 \tilde{v}\right) dx = \\
&= -\int_\Omega \operatorname{div}(T - \operatorname{div} K) \cdot \tilde{v} dx + \int_{\partial^{d-1}\Omega} [(T - \operatorname{div} K)n - \operatorname{div}_P((Kn)P)] \cdot \tilde{v} dx + \\
&\qquad + \int_{\partial^{d-1}\Omega} (Kn)n \cdot \frac{\partial \tilde{v}}{\partial n} dx + \int_{\partial^{d-2}\Omega} \{\!\{(Kn)\tau\}\!\} \cdot \tilde{v} dx.
\end{aligned}
$$

Moreover the remaining terms of the virtual power can be written as

$$
\begin{aligned}
-\int_\Omega &\operatorname{div} T_e : \tilde{v} dx + \int_{\partial^{d-1}\Omega} T_e n \cdot \tilde{v} dx + \int_\Omega (T_p - \operatorname{dev} T_e - \operatorname{div} K_p) : \tilde{V}_p dx + \\
&+ \int_{\partial^{d-1}\Omega} K_p n : \tilde{V}_p dx - \int_\Omega \frac{1}{d} \operatorname{tr} T_e \tilde{\mu} dx.
\end{aligned}
$$

Now the results follows from the fundamental lemma of the calculus of variations. □

3.1.3 Constitutive relations

We assume a constitutive relation for the free energy density of the form

$$\psi = \frac{1}{2}\mathfrak{C}E_e : E_e + \frac{1}{2}\mathfrak{A}M \mathbin{\vdots} M + \frac{1}{2}\mathfrak{B}\mathbb{U}_p \circ \mathbb{U}_p.$$

with $\mathfrak{C}$, $\mathfrak{A}$ and $\mathfrak{B}$ a symmetric and positive definite linear operators on Sym, Lin $\otimes\,\mathbb{R}$ and DevPla respectively. Moreover we choose a dissipation potential density of the form

$$\phi = \frac{1}{2}\mathfrak{D}\dot{E} : \dot{E} + \frac{1}{2}\mathfrak{E}\dot{M} \mathbin{\vdots} \dot{M} + \frac{1}{2}\mathfrak{F}\dot{\mathbb{U}}_p \circ \dot{\mathbb{U}}_p + Y(p)G(\dot{U}_p).$$

Here $\mathfrak{D}, \mathfrak{E}, \mathfrak{F}$ are symmetric and positive definite linear operators on Sym, Lin$\otimes\mathbb{R}$ and DevPla respectively. Moreover $G\colon \mathrm{DevSym} \to [0,\infty]$ is the convex and subdifferentiable plastic dissipation potential satisfying $G(0) = 0$ and $Y\colon \mathbb{R} \to [0,\infty)$ is the pressure dependent yield strength. Consistently with these choices of the free energy density and of the dissipation potential density we impose the following constitutive relations. The macroscopic response is described by

$$T = \mathfrak{D}\dot{E} \quad \text{and} \quad K = \mathfrak{A}M + \mathfrak{E}\dot{M}.$$

with $\mathfrak{D}$ and $\mathfrak{E}$ symmetric and positive definite linear operators on Sym and Lin $\otimes\,\mathbb{R}$ respectively. The elastic response is given by

$$T_e = \mathfrak{C}E_e$$

and the viscoplastic response by

$$\mathbb{T}_p \in \mathfrak{B}\mathbb{U}_p + \mathfrak{F}\dot{\mathbb{U}}_p + Y(p)\partial_{\dot{U}_p} G(\dot{U}_p)$$

where $\mathfrak{F}$ is a symmetric positive definite tensor on DevPla, $G\colon \mathrm{DevSym} \to \mathbb{R}$ is the plastic dissipation potential satisfying $\partial_{\dot{U}_p} G(\dot{U}_p) : \dot{U}_p \geqslant 0$ and $Y\colon \mathbb{R} \to [0,\infty)$ is the pressure dependent yield strength.

Remark 3.2. In the above expression we are viewing $Y(p)\partial_{\dot{U}_p} G(\dot{U}_p) \in \mathrm{DevSym}$ as an element of DevPla by considering DevSym as a subspace of DevPla.

3.1.4 Refinements of the model

To include the dilatancy in the model we could do the following. We put the unilateral constrain $\lambda_p \geqslant F(U_p)$, meaning that when the microscopic structure suffer shear distortions it must suffer also an expansion. This constrain is still imposed through the Lagrange multiplier p representing the microscopic pressure. To make the expansion irreversible (once the microscopic structure expanded it can not be compacted again) we add a microscopic force ξ conjugate to $\tilde{\mu}_p$ in the internal virtual power and in the dissipation potential density we add $\iota_{[0,\infty)}(\dot{\xi})$. In this way ξ opposes to the compaction of the microscopic structure.

3.2 Mathematical analysis

In what follows we write simply U and λ in place of U_p and λ_p respectively. Moreover for simplicity we assume $\Gamma_D = \partial\Omega$ and $\overline{u} = 0$ since the extension to the general case in trivial.

3.2.1 Hypotheses

To get the global existence result we assume the following hypotheses.

(S) (a) $\Omega \subset \mathbb{R}^d$ with $d \in \{2, 3\}$ is open, bounded and Lipschitz
(b) Γ_D and Γ_N are relatively open disjoint subsets of $\partial\Omega$

(Y) (a) $Y \in C^{0,1}(\mathbb{R}, [0, \infty))$

(G) (a) $G \in C^{1,1}(\mathrm{DevSym}, \mathbb{R})$ is convex
(b) $\partial G \in L^\infty(\mathrm{DevSym}, \mathrm{DevPla})$
(c) $\partial G(\dot{U}) : \dot{U} \geqslant 0$ for all $\dot{U} \in \mathrm{DevSym}$

(L) (a) $f \in C^1([0, T], L^2(\Omega)^d)$
(b) $h \in C^1([0, T], L^2(\Gamma_N)^d)$

(E) (a) $\mathfrak{C}$ and $\mathfrak{D}$ are in $\mathrm{Sym} \otimes \mathrm{Sym}$, $\mathfrak{B}$ and $\mathfrak{F}$ are in $\mathrm{DevPla} \otimes \mathrm{DevPla}$, $\mathfrak{A}$ and $\mathfrak{E}$ are in $(\mathrm{Lin} \otimes \mathbb{R}) \otimes (\mathrm{Lin} \otimes \mathbb{R})$
(b) there exist $\gamma > 0$ and $\delta > 0$ such that $\mathfrak{C}E : E \geqslant \gamma|E|^2$ and $\mathfrak{D}E : E \geqslant \delta|E|^2$ for all $E \in \mathrm{Sym}$, there exist $\beta > 0$ and ϕ such that $\mathfrak{B}\mathbb{U} \circ \mathbb{U} \geqslant \beta|\mathbb{U}|^2$ and $\mathfrak{F}\mathbb{U} \circ \mathbb{U} \geqslant \phi|\mathbb{U}|^2$, there exists $\alpha > 0$ and $\epsilon > 0$ such that $\mathfrak{A}M \vdots M \geqslant \alpha|\operatorname{sym} M|^2$ and $\mathfrak{E}M \vdots M \geqslant \epsilon|\operatorname{sym} M|^2$ for all $M \in \mathrm{Lin} \otimes \mathbb{R}$ where sym denote the symmetrization of the first two indices

3.2.2 Function spaces

We now introduce the function spaces that we use to formulate the problem. Let

$$\mathcal{Z} = \left\{ y = (u, U, \lambda) \middle| \begin{array}{c} u \in H^2(\Omega)^d \text{ and } u = 0 \text{ on } \Gamma_D \\ U \in H^1(\Omega, \mathrm{DevSym}) \\ \lambda \in L^2(\Omega) \end{array} \right\} \quad \text{and} \quad \mathcal{Q} = \left\{ p \in L^2(\Omega) \right\}.$$

The rates of $y = (u, U, \lambda) \in \mathcal{Z}$ will be denoted by $\dot{y} = (\dot{u}, \dot{U}, \dot{\lambda}) \in \mathcal{Z}$. The variations of $\dot{y} = (\dot{u}, \dot{\xi}, \dot{\lambda}) \in \mathcal{Z}$ and $p \in \mathcal{Q}$ will be denoted as $z = (v, V, \mu) \in \mathcal{Z}$ and $q \in \mathcal{Q}$. The test functions will be denoted as $\tilde{z} = (\tilde{v}, \tilde{V}, \tilde{\mu})$.

3.2.3 Formulation of the problem

We define $\mathcal{F} \colon \mathcal{Z} \times \mathcal{Q} \times \mathcal{Z} \times [0, T] \to \mathcal{Z}' \times \mathcal{Q}'$ by

$$\begin{aligned} \langle \mathcal{F}(\dot{y}, p, y, t), (\tilde{z}, \tilde{q}) \rangle = \int_\Omega \mathfrak{C} \left(Eu - U - \frac{\lambda}{d} I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \\ + \int_\Omega (\mathfrak{A} \operatorname{grad}^2 u + \mathfrak{E} \operatorname{grad}^2 \dot{u}) \vdots \operatorname{grad}^2 \tilde{v} dx + \int_\Omega \mathfrak{D} E\dot{u} : E\tilde{v} dx + \\ + \int_\Omega \left[(\mathfrak{B}\mathbb{U} + \mathfrak{F}\dot{\mathbb{U}}) \circ \tilde{\mathbb{V}} + Y(p) \partial G(\dot{U}) : \tilde{V} \right] dx - \\ - \int_\Omega p\tilde{\mu} dx - \int_\Omega \lambda \tilde{q} dx - \\ - \int_\Omega f(t) \cdot \tilde{v} dx - \int_{\Gamma_N} h(t) \cdot \tilde{v} dx. \end{aligned}$$

Lemma 3.3. The map $\mathcal{F}$ is well defined.

Proof. The linear terms are trivially well defined so we need to check only

$$\int_\Omega Y(p)\partial G(\dot{\mathbb{U}}) : \tilde{\mathbb{V}}dx. \tag{3.2.1}$$

But

$$\int_\Omega |Y(p)||\partial G(\dot{\mathbb{U}})||\tilde{\mathbb{V}}|dx \leqslant \|\partial G\|_\infty \int_\Omega (Y(0) + \mathrm{Lip}(Y)|p|)|\tilde{\mathbb{V}}|dx \lesssim (1 + \|p\|_2)\|\tilde{V}\|_{1,2}.$$

This concludes the proof. □

Now we can formulate the evolution problem as follows.

Problem 3.4. Find $y \in H^1((0,T),\mathcal{Z})$ and $p \in L^2([0,T],\mathcal{Q})$ such that $y(0) = 0$ and

$$\mathcal{F}(\dot{y}(t), p(t), y(t), t) = 0$$

for almost every $t \in (0,T)$.

3.2.4 Global existence

This section is devoted to prove the following existence of a global in time solution to Problem 3.4.

Theorem 3.5. Assume the hypotheses of Section 3.2.1. Then there exists $y \in H^1((0,T),\mathcal{Z})$ and $p \in C^0([0,T],\mathcal{Q})$ solution to Problem 3.4.

The proof is based on the Rothe method so we first prove the existence to a discrete in time version of the problem and then we pass to the vanishing time step limit.

Existence of the discrete in time solution

Let $N \in \mathbb{N}$. We define $\tau_N = N^{-1}T$ and $t_n^N = n\tau_N$ for $n \in \{0,\dots,N\}$. We consider the discrete in time version of Problem 3.4.

Problem 3.6 (Discrete problem)**.** Let $y_0^N = 0 \in \mathcal{Z}$. Find $y_n^N \in \mathcal{Z}$ and $p_n^N \in \mathcal{Q}$ for $n \in \{1,\dots,N\}$ such that

$$\mathcal{F}\left(\frac{y_n^N - y_{n-1}^N}{\tau_N}, p_n^N, y_n^N, t_n^N\right) = 0 \tag{3.2.2}$$

for all $n \in \{1,\dots,N\}$.

We first prove the existence of a solution to Problem 3.6. Reasoning by induction we reduce to find $y_n^N \in \mathcal{Z}$ and $p_n^N \in \mathcal{Q}$ that satisfy (3.2.2) given $y_{n-1}^N \in \mathcal{Z}$. We immediately observe that in view of the constrain $\lambda_n^N = 0$ for all n. To simplify the notation we call $y = y_{n-1}^N$, $\dot{y} = \tau^{-1}(y_n^N - y_{n-1}^N)$ and $t = t_n^N$. Then $y_n^N = y + \tau\dot{y}$. It is enough to prove that there exists $\dot{y} \in \mathcal{Z}$ and $p \in Q$ such that

$$\mathcal{F}(\dot{y}, p, y + h\dot{y}, t) = 0. \tag{3.2.3}$$

We define the operator $\mathcal{A} : \mathcal{Q} \times \mathcal{Z} \to \mathcal{Z}'$ by

$$\langle \mathcal{A}(p,\dot{y}), \tilde{z}\rangle = \int_\Omega \tau\mathfrak{C}\left(E\dot{u} - \dot{\mathrm{U}} - \frac{\dot{\lambda}}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right)dx + \int_\Omega \mathfrak{D}E\dot{u} : E\tilde{v}dx +$$

$$+ \int_\Omega (\mathfrak{E} + \tau\mathfrak{A})\,\mathrm{grad}^2\,\dot{u} \,\vdots\, \mathrm{grad}^2\,\tilde{v}dx + \int_\Omega \left[(\mathfrak{F} + \tau\mathfrak{B})\dot{\mathbb{U}} \circ \tilde{\mathbb{V}} + Y(p)\partial G(\dot{\mathrm{U}}) : \tilde{V}\right]dx$$

the linear operator $\mathcal{B}\colon \mathcal{Q} \to \mathcal{Z}'$ by

$$\langle \mathcal{B}p, \tilde{z}\rangle = -\int_\Omega p\tilde{\mu}dx$$

and the operator $\mathcal{C}\colon \mathcal{Z} \to \mathcal{Q}'$ by

$$\langle \mathcal{C}(\dot{y}), \tilde{q}\rangle = \int_\Omega (\lambda + \tau\dot{\lambda})\tilde{q}\,dx = 0.$$

Moreover we introduce the linear form $l \in \mathcal{Z}'$

$$\begin{aligned}\langle l, \tilde{z}\rangle = \int_\Omega f(t)\cdot\tilde{v}dx + \int_{\Gamma_N} h(t)\cdot\tilde{v}dx-\\ -\int_\Omega \mathfrak{C}\left(Eu - U - \frac{\lambda}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right)dx-\\ -\int_\Omega \mathfrak{A}\,\mathrm{grad}^2 u \,\vdots\, \mathrm{grad}^2\tilde{v}dx - \int_\Omega \mathfrak{B}\mathbb{U}\circ\tilde{\mathbb{V}}dx.\end{aligned}$$

Now equation (3.2.3) can be written as

$$\begin{cases}\mathcal{A}(p,\dot{y}) + Bp = l(t)\\ \mathcal{C}(\dot{y}) = 0.\end{cases}$$

We have already noticed that $\lambda = 0$ and $\dot{\lambda} = 0$. We call

$$\mathcal{N} = \left\{\dot{y} = (\dot{u}, \dot{U}, 0)\;\middle|\; \substack{\dot{u}\in H^2(\Omega)^d \text{ and } \dot{u}=0 \text{ on } \Gamma_D\\ \dot{U}\in H^1(\Omega,\mathrm{DevSym})}\right\}.$$

Now we study the first equation in (3.2.3). We first study the solvability of that equation in $\dot{y} \in \mathcal{N}$ given $p \in \mathcal{Q}$. If we test the first equation with $\tilde{z} \in \mathcal{N}$ we get

$$\langle \mathcal{A}(p,\dot{y}), \tilde{z}\rangle = \langle l, \tilde{z}\rangle \quad \text{for all } \tilde{z}\in\mathcal{N}.$$

Now we define the space

$$\mathcal{X} = \left\{\xi = (\dot{u}, \dot{U})\;\middle|\; \substack{\dot{u}\in H^2(\Omega)^d \text{ and } \dot{u}=0 \text{ on } \Gamma_D\\ \dot{U}\in H^1(\Omega,\mathrm{DevSym})}\right\}$$

and we can consider the natural bijection $\mathcal{J}$ from $\mathcal{X}$ to $\mathcal{N}$. In particular if $\xi = (\dot{u}, \dot{U})$ we define

$$\mathcal{J}(\xi) = (\dot{u}, \dot{U}, 0).$$

Now we introduce the operator $\mathcal{T}_p$ from $\mathcal{X}$ to $\mathcal{X}'$ as follows. Given $\xi = (\dot{u}, \dot{U})$ and $\tilde{\zeta} = (\tilde{v}, \tilde{V})$ in $\mathcal{X}$ we set

$$\begin{aligned}\langle \mathcal{T}_p(\xi), \tilde{\zeta}\rangle = \langle \mathcal{A}(p, \mathcal{J}(\xi)), \mathcal{J}(\tilde{\zeta})\rangle = \int_\Omega \tau\mathfrak{C}\left(E\dot{u} - \dot{U}\right) : \left(E\tilde{v} - \tilde{V}\right)dx + \int_\Omega \mathfrak{D}E\dot{u} : E\tilde{v}dx+\\ +\int_\Omega (\mathfrak{E} + \tau\mathfrak{A})\,\mathrm{grad}^2\dot{u} \,\vdots\, \mathrm{grad}^2\tilde{v}dx + \int_\Omega \left[(\mathfrak{F} + \tau\mathfrak{B})\dot{\mathbb{U}}\circ\tilde{\mathbb{V}} + Y(p)\partial G(\dot{U}) : \tilde{V}\right]dx.\end{aligned}$$

Now for each $p \in \mathcal{Q}$ we are reduced to find $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\zeta}\rangle = \langle l, \mathcal{J}(\tilde{\zeta})\rangle \quad \text{for all } \tilde{\zeta}\in\mathcal{X}. \tag{3.2.4}$$

We want to apply the Browder–Minty theorem (Theorem C.16). We need to establish the boundedness, continuity, coercivity and monotonicity of $\mathcal{T}_p$. This done in the following lemmas.

Lemma 3.7. The operator $\mathcal{T}_p : \mathcal{X} \to \mathcal{X}'$ is bounded and continuous.

Proof. We have

$$|\langle \mathcal{T}_p(\xi), \tilde{\zeta}\rangle| \lesssim (\|E\dot{u}\|_2 + \|\dot{U}\|_2)(\|E\tilde{v}\|_2 + \|\tilde{V}\|_2) + \\ + \|\mathrm{grad}^2\,\dot{u}\|_2 \|\mathrm{grad}^2\,\tilde{v}\|_2 + \|\dot{\mathbb{U}}\|_2 \|\tilde{\mathbb{V}}\|_2 + (1 + \|p\|_2)\|\tilde{\mathbb{V}}\|_2.$$

and this proves that $\mathcal{T}_p$ is bounded. Now it is enough to prove the continuity of the nonlinear term only. Let $\xi_n \to \xi$ in $\mathcal{X}$. Then $\dot{\mathbb{U}}_n \to \dot{\mathbb{U}}$ in L^2. We have

$$\left| \int_\Omega Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}})) \circ \tilde{\mathbb{V}} dx \right| \leqslant \|Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))\|_2 \|\tilde{\mathbb{V}}\|_2$$

and

$$\|Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))\|_2^2 = \int_\Omega Y(p)^2 (\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))^2 dx \to 0.$$

Indeed for every subsequence there exists a further subsequence such that $\dot{\mathbb{U}}_n \to \dot{\mathbb{U}}$ in L^2 and we can apply the dominated convergence theorem to pass to the limit. □

Lemma 3.8. The operator $\mathcal{T}_p : \mathcal{X} \to \mathcal{X}'$ is coercive uniformly in p.

Proof. For any $\xi \in \mathcal{X}$ we have

$$\langle \mathcal{T}_p(\xi), \xi\rangle = \int_\Omega \tau \mathfrak{C} \left(E\dot{u} - \dot{U}\right) : \left(E\dot{u} - \dot{U}\right) dx + \int_\Omega \mathfrak{D} E\dot{u} : E\dot{u} dx + \\ + \int_\Omega (\mathfrak{E} + \tau\mathfrak{A})\, \mathrm{grad}^2\, \dot{u} \,\vdots\, \mathrm{grad}^2\, \dot{u} dx + \int_\Omega \left[(\mathfrak{F} + \tau\mathfrak{B})\dot{\mathbb{U}} \circ \dot{\mathbb{U}} + Y(p)\partial G(\dot{U}) \circ \dot{U}\right] dx.$$

Using the hypotheses (E, b) and (G, b) and using that $\dot{U} : I = 0$ since $\dot{U} \in \mathrm{Dev}$ we have

$$\langle \mathcal{T}_p(\xi), \xi\rangle \geqslant \tau\gamma \left\|E\dot{u} - \dot{U}\right\|_2^2 + \tau\alpha \|\mathrm{grad}\, E\dot{u}\|_2^2 + (\phi + \tau\beta)\|\dot{\mathbb{U}}\|_2^2.$$

By the Young inequality

$$\left\|E\dot{u} - \dot{U}\right\|_2^2 \geqslant (1 - \theta)\|E\dot{u}\|_2^2 + \left(1 - \frac{1}{\theta}\right)\|\dot{U}\|_2^2$$

for any $\theta > 0$. Then

$$\langle \mathcal{T}_p(\xi), \xi\rangle \geqslant (1 - \theta)\tau\gamma\|E\dot{u}\|_2^2 + \tau\alpha\|\mathrm{grad}\, E\dot{u}\|_2^2 + \\ + \left(1 - \frac{1}{\theta}\right)\tau\gamma\|\dot{U}\|_2^2 + (\phi + \tau\beta)\|\dot{\mathbb{U}}\|_2^2.$$

We choose $\theta < 1$ sufficiently close to 1 such there exists a constant $C_1 > 0$ with the property that

$$\left(1 - \frac{1}{\theta}\right)\tau\gamma\|\dot{U}\|_2^2 + (\phi + \tau\beta)\|\dot{\mathbb{U}}\|_2^2 \geqslant C_1\|\dot{\mathbb{U}}\|_2^2 \geqslant C_2\|\dot{U}\|_{1,2}^2.$$

The thesis follows by application of Korn inequality. □

Lemma 3.9. The operator $\mathcal{T}_p : \mathcal{X} \to \mathcal{X}'$ is strongly monotone uniformly in p, i.e., there exists $M > 0$ such that

$$\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1\rangle \geqslant M\|\xi_2 - \xi_1\|_{\mathcal{X}}^2$$

for all $p \in \mathcal{Q}$ and ξ_1, ξ_2 in $\mathcal{X}$.

Proof. For all ξ_1 and ξ_2 in $\mathcal{X}$ we have

$$
\begin{aligned}
&\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle = \\
&= \int_\Omega \tau\mathfrak{C}\left(E(\dot{u}_2 - \dot{u}_1) - (\dot{U}_2 - \dot{U}_1)\right) : (E(\dot{u}_2 - \dot{u}_1) - (U_2 - U_1))\,dx + \\
&+ \int_\Omega \mathfrak{D}(E(\dot{u}_2 - \dot{u}_1)) : (E(\dot{u}_2 - \dot{u}_1))dx + \\
&+ \int_\Omega (\mathfrak{E} + \tau\mathfrak{A})(\operatorname{grad}^2 \dot{u}_2 - \operatorname{grad}^2 \dot{u}_1) \vdots (\operatorname{grad}^2 \dot{u}_2 - \operatorname{grad}^2 \dot{u}_1)dx + \\
&+ \int_\Omega \left[(\mathfrak{F} + \tau\mathfrak{B})(\dot{\mathbb{U}}_2 - \dot{\mathbb{U}}_1) \circ (\dot{\mathbb{U}}_2 - \dot{\mathbb{U}}_1) + Y(p)(\partial G(\dot{U}_2) - \partial G(\dot{U}_1)) : (\dot{U}_2 - \dot{U}_1)\right] dx \geqslant \\
&\geqslant (\delta + \tau\gamma)\| E(\dot{u}_2 - \dot{u}_1) - (\dot{U}_2 - \dot{U}_1)\|_2^2 + (\phi + \tau\beta)\|\dot{\mathbb{U}}_2 - \dot{\mathbb{U}}_1\|_2^2.
\end{aligned}
$$

We have used that since G is convex then ∂G is monotone. Reasoning as in the proof of Lemma 3.8 we have

$$
\begin{aligned}
\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle \geqslant (1-\theta)(\delta + \tau\gamma)\| E(\dot{u}_2 - \dot{u}_1)\|_2^2 + (\alpha + \tau\epsilon)\|\operatorname{grad} E(\dot{u}_2 - \dot{u}_1)\|_2^2 + \\
+ \left(1 - \frac{1}{\theta}\right)(\delta + \tau\gamma)\|\dot{U}_2 - \dot{U}_1\|_2^2 + \\
+ (\phi + \tau\beta)\|\dot{\mathbb{U}}_2 - \dot{\mathbb{U}}_1\|_2^2.
\end{aligned}
$$

Choosing $0 < \theta < 1$ sufficiently close to 1 and using second order Korn inequality we conclude. □

Now applying Theorem C.16 for each $p \in \mathcal{Q}$ there exists $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\zeta} \rangle = \langle l, \mathcal{J}(\tilde{\zeta}) \rangle \quad \text{for all } \tilde{\zeta} \in \mathcal{X}.$$

Moreover ξ is unique. Indeed by strong monotonicity if ξ_1 and ξ_2 are two solutions we have

$$M\|\xi_2 - \xi_1\|_{\mathcal{X}}^2 \leqslant \langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle = 0.$$

We call $\Xi\colon \mathcal{Q} \to \mathcal{X}$ the map that assign to $p \in \mathcal{Q}$ the unique ξ. Now to conclude the existence of a solution to Problem 3.6 we need to solve in $p \in \mathcal{Q}$ the equation

$$\langle \mathcal{A}(p, \mathcal{J}(\Xi(p))), (0,0,\tilde{\mu}) \rangle + \langle Bp, (0,0,\tilde{\mu}) \rangle = \langle l, (0,0,\tilde{\mu}) \rangle \quad \text{for all } \tilde{\mu} \in L^2(\Omega)$$

that is

$$\int_\Omega \operatorname{tr}\left[\tau\mathfrak{C}\left(E\dot{u} - \dot{U}\right)\right]\frac{\tilde{\mu}}{d}dx - \int_\Omega p\tilde{\mu}dx = \int_\Omega \operatorname{tr}\left[\mathfrak{C}\left(Eu - U\right)\right]\frac{\tilde{\mu}}{d}dx$$

for all $\tilde{\mu} \in L^2(\Omega)$, where $\dot{y} = (\dot{u}, \dot{U}, \dot{\lambda}) = \mathcal{J}(\Xi(p))$. This is equivalent to

$$p = \frac{1}{d}\operatorname{tr}\left[\tau\mathfrak{C}\left(E\dot{u} - \dot{U}\right)\right] - \frac{1}{d}\operatorname{tr}\left[\mathfrak{C}\left(Eu - U\right)\right] = \mathcal{S}(p) \tag{3.2.5}$$

where we have introduced the operator $\mathcal{S}\colon \mathcal{Q} \to \mathcal{Q}$. We want to prove the existence of a fixed point of $\mathcal{S}$ using the Schauder fixed point theorem (Theorem C.11). To prove that $\mathcal{S}$ satisfy the required hypotheses we need to establish some regularity results regarding Ξ. We first need the following lemma.

Lemma 3.10. The operator $\mathcal{A}$ is Lipschitz continuous in p uniformly in $\dot{y}$.

Proof. By (Y, a) and (G, c) we have

$$
\begin{aligned}
|\langle \mathcal{A}(p,\dot{y}) - \mathcal{A}(q,\dot{y}), \tilde{z}\rangle| &\leqslant \int_\Omega |Y(p) - Y(q)||\partial G(\dot{U})||\tilde{V}|\,dx \leqslant \\
&\leqslant \mathrm{Lip}(Y)\|\partial G\|_\infty \int_\Omega |p-q||\tilde{V}|\,dx \leqslant L\|p-q\|_2\|\tilde{V}\|_2
\end{aligned}
$$

for some constant $L > 0$. Then we have

$$\|\mathcal{A}(p,\dot{y}) - \mathcal{A}(q,\dot{y})\|_{\mathcal{Z}'} \leqslant L\|p-q\|_{\mathcal{Q}}$$

This concludes the proof. □

Now we can prove the continuity of Ξ.

Lemma 3.11. The operator Ξ is continuous and has bounded range.

Proof. First we prove that Ξ has bounded range. We assume by contradiction that there exists $p_n \in \mathcal{Q}$ such that $\|\Xi(p_n)\|_{\mathcal{X}} \geqslant n$. Let $\xi_n = \Xi(p_n)$. Now

$$\frac{\langle \mathcal{T}_{p_n}(\xi_n), \xi_n\rangle}{\|\xi_n\|_{\mathcal{X}}} \leqslant \|\langle l, \mathcal{J}(\cdot)\rangle\|_{\mathcal{X}'}.$$

But by Lemma 3.8

$$\frac{\langle \mathcal{T}_p(\xi_n), \xi_n\rangle}{\|\xi_n\|_{\mathcal{X}}} \to \infty$$

and this is a contradiction. Now we prove that Ξ is Lipschitz continuous. Indeed let $p_i \in \mathcal{Q}$ and $\xi_i = \Xi(p_i)$ with i equals to 1 or 2. Since $\mathcal{T}_{p_1}(\xi_1) = \mathcal{T}_{p_2}(\xi_2)$, $\mathcal{T}_p$ is strictly monotone uniformly in p and $\mathcal{A}$ is uniformly Lipschitz in p we have

$$
\begin{aligned}
M\|\xi_2 - \xi_1\|^2 &\leqslant \langle \mathcal{T}_{p_1}(\xi_2) - \mathcal{T}_{p_1}(\xi_1), \xi_2 - \xi_1\rangle = \langle \mathcal{T}_{p_1}(\xi_2) - \mathcal{T}_{p_2}(\xi_2), \xi_2 - \xi_1\rangle \leqslant \\
&\leqslant L\|p_2 - p_1\|_{\mathcal{Q}}\|\xi_2 - \xi_1\|_{\mathcal{X}}.
\end{aligned}
$$

This concludes the proof. □

The following lemma establish the regularity of $\mathcal{J}$.

Lemma 3.12. The map $\mathcal{J}\colon \mathcal{X} \to \mathcal{Z}$ is continuous and bounded.

Proof. It is trivial since $\mathcal{J}(\xi) = (\dot{u}, \dot{U}, 0)$. □

Now we are ready to prove the desired properties of $\mathcal{S}$.

Lemma 3.13. The map $\mathcal{S}\colon \mathcal{Q} \to \mathcal{Q}$ is continuous and has a bounded and compact range.

Proof. We notice that since Ξ is continuous and has compact range then also $\mathcal{S}$ is continuous and has compact range. We notice that by Rellich–Kondarchov theorem then $\dot{u} \mapsto E\dot{u}$ from H^2 to L^2 and $\dot{U} \mapsto \dot{U}$ from H^1 to L^2 are compact. Then $\mathcal{S}$ is compact. Since $\mathcal{S}$ has bounded range and is compact then it has compact range. □

Now we are in position to apply Schauder fixed point theorem and we get the existence of a fixed point of $\mathcal{S}$. We have obtained the desired result.

Proposition 3.14. Problem 3.6 admits at least a solution.

Now we redefine the operator $\mathcal{A}\colon \mathcal{Z} \to \mathcal{Z}'$ by

$$\langle \mathcal{A}y, \tilde{z}\rangle = \int_\Omega \mathfrak{C}\left(Eu - U - \frac{\lambda}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right) dx + \int_\Omega \mathfrak{A}\operatorname{grad} u \,\vdots\, \operatorname{grad}\tilde{v} dx + \int_\Omega \mathfrak{B}\mathbb{U} \circ \tilde{\mathbb{V}} dx$$

and we define operator $\mathcal{E}\colon \mathcal{Z} \to \mathcal{Z}'$ by

$$\langle \mathcal{E}\dot{y}, \tilde{z}\rangle = \int_\Omega \mathfrak{D}E\dot{u} : E\tilde{v} dx + \int_\Omega \mathfrak{E}\operatorname{grad}\dot{u} \,\vdots\, \operatorname{grad}\tilde{v} dx + \int_\Omega \mathfrak{F}\dot{\mathbb{U}} \circ \tilde{\mathbb{V}} dx.$$

We define also $\mathcal{D}\colon \mathcal{Q} \times \mathcal{Z} \to \tilde{\mathcal{Z}}'$ by

$$\langle \mathcal{D}(p, \dot{y}), \tilde{z}\rangle = \int_\Omega Y(p)\partial G(\dot{U}) : \tilde{V} dx.$$

The operator $\mathcal{B}$ is defined exactly as before and and $\mathcal{C}\colon \mathcal{Z} \to \mathcal{Q}'$ is redefined by

$$\langle \mathcal{C}y, \tilde{q}\rangle = \int_\Omega \lambda\tilde{q} dx = 0.$$

Let also $l\colon [0, T] \to \mathcal{Z}'$ redefined by

$$\langle l(t), \tilde{z}\rangle = \int_\Omega f(t) \cdot \tilde{v} dx + \int_{\Gamma_N} h(t) \cdot \tilde{v} dx.$$

Proposition 3.14 ensures that that for each N there exists $\left\{(\dot{y}^N_n, p^N_n, y^N_n)\right\}^N_{n=0}$ such that $(\dot{y}^N_0, p^N_0, y^N_0) = 0$, $y^N_n = y^N_{n-1} + \tau\dot{y}^N_n$ and

$$\begin{cases} \mathcal{A}y^N_n + \mathcal{E}\dot{y}^N_n + \mathcal{D}(p^N_n, \dot{y}^N_n) + \mathcal{B}p^N_n = l^N_n \\ \mathcal{C}y^N_n = 0 \end{cases} \tag{3.2.6}$$

for all $n \in \{1, \dots, N\}$. Here $l^N_n = l(t^N_n)$.

Construction of the continuous in time solution

Now we want to construct the continuous in time solution passing to the limit $N \to \infty$. We define

$$\begin{aligned} l^N &\in L^2((0,T), \mathcal{Z}') && \text{piecewise constant interpolation of } \{l^N_n\}^N_{n=1} \\ \overline{y}^N &\in L^2((0,T), \mathcal{Z}) && \text{piecewise constant interpolation of } \{y^N_n\}^N_{n=1} \\ y^N &\in H^1((0,T), \mathcal{Z}) && \text{piecewise linear interpolation of } \{y^N_n\}^N_{n=1} \\ \dot{y}^N &\in L^2((0,T), \mathcal{Z}) && \text{piecewise constant interpolation of } \{\dot{y}^N_n\}^N_{n=1} \\ p^N &\in L^2((0,T), \mathcal{Q}) && \text{piecewise constant interpolation of } \{p^N_n\}^N_{n=1}. \end{aligned}$$

We will need the following lemma.

Lemma 3.15. We have that $l^N \to l$ in $L^2((0,T), \mathcal{Z}')$ and in particular

$$\|l^N\|^2_{L^2((0,T),\mathcal{Z}')} = \sum_{n=1}^N \tau_N \|l^N_n\|^2_{\mathcal{Z}'}$$

is bounded.

Proof. For all $t \in [t^N_{n-1}, t^N_n]$ we have

$$l^N(t) = l^N_n = l(t^N_n) = l(t) + \int_t^{t^N_n} \dot{l}(s)ds.$$

By Hölder inequality

$$\|l^N(t) - l(t)\|_{\mathcal{Z}'} \leqslant \int_t^{t^N_n} \|\dot{l}(s)\|_{\mathcal{Z}'}ds \leqslant \int_{t^N_{n-1}}^{t^N_n} \|\dot{l}(s)\|_{\mathcal{Z}'}ds \leqslant \sqrt{\tau_N}\left[\int_{t^N_{n-1}}^{t^N_n} \|\dot{l}(s)\|^2_{\mathcal{Z}'}ds\right]^{1/2}.$$

Taking the square and integrating for $t \in [t^N_{n-1}, t^N_n]$ we have

$$\int_{t^N_{n-1}}^{t^N_n} \|l^N(t) - l(t)\|^2_{\mathcal{Z}'}dt \leqslant \tau_N^2 \int_{t^N_{n-1}}^{t^N_n} \|\dot{l}(s)\|^2_{\mathcal{Z}'}ds.$$

Adding this inequalities for $n \in \{1, \dots, N\}$ we have

$$\|l^N - l\|^2_{L^2((0,T),\mathcal{Z}')} \leqslant \tau_N^2\|\dot{l}\|^2_{L^2((0,T),\mathcal{Z}')} \to 0.$$

This concludes the proof. □

We now obtain a priori estimates on $\overline{y}^N$, y^N and $\dot{y}^N$.

Lemma 3.16. The functions $\overline{y}^N$ and y^N are bounded in $L^\infty((0,T),\mathcal{Z})$ and the function $\dot{y}^N$ is bounded in $L^2((0,T),\mathcal{Z})$.

Proof. We first observe that y^N_n and $\dot{y}^N_n$ belongs to $\mathcal{N}$ and that $\mathcal{A}$ and $\mathcal{E}$ are coercive there. Next we test the first equation in (3.2.6) with $\tilde{z} = \dot{y}^N_n$. We notice that $\langle \mathcal{D}(p^N_n, \dot{y}^N_n), \dot{y}^N_n\rangle \geqslant 0$ and $\langle \mathcal{B}p^N_n, \dot{y}^N_n\rangle = 0$. Then

$$\frac{1}{\tau_N}\langle \mathcal{A}y^N_n, y^N_n - y^N_{n-1}\rangle + \langle \mathcal{E}\dot{y}^N_n, \dot{y}^N_n\rangle \leqslant \langle l^N_n, \dot{y}^N_n\rangle.$$

Multiplying by $2\tau_N$ we have

$$2\langle \mathcal{A}y^N_n, y^N_n - y^N_{n-1}\rangle + 2\tau_N\langle \mathcal{E}\dot{y}^N_n, \dot{y}^N_n\rangle \leqslant 2\tau_N\langle l^N_n, \dot{y}^N_n\rangle.$$

Now

$$\begin{aligned} 2\langle \mathcal{A}y^N_n, y^N_n - y^N_{n-1}\rangle &= \langle \mathcal{A}y^N_n, y^N_n\rangle - \langle \mathcal{A}y^N_{n-1}, y^N_{n-1}\rangle + \langle \mathcal{A}(y^N_n - y^N_{n-1}), (y^N_n - y^N_{n-1})\rangle \geqslant \\ &\geqslant \langle \mathcal{A}y^N_n, y^N_n\rangle - \langle \mathcal{A}y^N_{n-1}, y^N_{n-1}\rangle. \end{aligned}$$

Let $E > 0$ be its coercivity constant of $\mathcal{E}$ in $\mathcal{N}$. Then

$$2\tau_N\langle \mathcal{E}\dot{y}^N_n, \dot{y}^N_n\rangle \geqslant 2\tau_N E\|\dot{y}^N_n\|^2_{\mathcal{Z}}$$

and using Young inequality

$$2\tau_N\langle l^N_n, \dot{y}^N_n\rangle \leqslant \frac{\tau_N}{E}\|l^N_n\|^2_{\mathcal{Z}'} + \tau_N E\|\dot{y}^N_n\|^2_{\mathcal{Z}}.$$

Then

$$\langle \mathcal{A}y^N_n, y^N_n\rangle - \langle \mathcal{A}y^N_{n-1}, y^N_{n-1}\rangle + \tau_N E\|\dot{y}^N_n\|^2_{\mathcal{Z}} \leqslant \frac{\tau_N}{E}\|l^N_n\|^2_{\mathcal{Z}'}. \tag{3.2.7}$$

Let $A > 0$ the coercivity constant of $\mathcal{A}$ on $\mathcal{N}$. Then adding the inequalities (3.2.7) we have

$$A\|y_n^N\|_{\mathcal{Z}}^2 + E\sum_{k=1}^{n} \tau_N \|\dot{y}_k^N\|_{\mathcal{Z}}^2 \leqslant \langle \mathcal{A}y_n^N, y_n^N\rangle + E\sum_{k=1}^{n} \tau_N \|\dot{y}_k^N\|_{\mathcal{Z}}^2 \leqslant \frac{1}{E}\sum_{k=1}^{n} \tau_N \|l_k^N\|_{\mathcal{Z}'}^2.$$

Using Lemma 3.15 we have that

$$\|y_n^N\|_{\mathcal{Z}} \quad \text{and} \quad \sum_{n=1}^{N} \tau_N \|\dot{y}_n^N\|_{\mathcal{Z}}^2$$

are bounded uniformly in $N \in \mathbb{N}$ and $n \in \{0, \dots, N\}$. This implies immediately the thesis. □

Using that lemma, up to extracting a subsequence we have that

$$\overline{y}^N \overset{*}{\rightharpoonup} \overline{y} \quad \text{in } L^\infty((0,T),\mathcal{Z}) \tag{3.2.8a}$$
$$y^N \overset{*}{\rightharpoonup} y \quad \text{in } L^\infty((0,T),\mathcal{Z}) \tag{3.2.8b}$$
$$\dot{y}^N \rightharpoonup z \quad \text{in } L^2((0,T),\mathcal{Z}). \tag{3.2.8c}$$

We notice that from $\lambda_n^N = 0$ it follows immediately that $\lambda = 0$.

Lemma 3.17. We have that $\overline{y} = y \in H^1((0,T),\mathcal{Z})$ and $\dot{y} = z$. Moreover $y(0) = 0$.

Proof. We prove that $\overline{y}^N - y^N \to 0$ in $L^\infty((0,T),\mathcal{Z})$. Indeed let $t \in [t_{n-1}^N, t_n^N]$. Then

$$\|\overline{y}^N(t_n^N) - y(t_n^N)\|_{\mathcal{Z}} = \|y_n^N - y_n^N - (t_n^N - t)\dot{y}_n^N\|_{\mathcal{Z}} \leqslant \tau_N \|\dot{y}_n^N\|_{\mathcal{Z}}.$$

It follows that

$$\|\overline{y}^N(t_n^N) - y(t_n^N)\|_{\mathcal{Z}}^2 \leqslant \tau_N^2 \|\dot{y}_n^N\|_{\mathcal{Z}}^2 \leqslant \tau_N \|\dot{y}^N\|_{L^2((0,T),\mathcal{Z})}^2 \to 0.$$

This establish the equality $\overline{y} = y$. Now let $\phi \in C_c^\infty([0,T))$ and $\zeta \in \mathcal{Z}'$ arbitrary. Then

$$\int_0^T \phi\langle \zeta, \dot{y}^N\rangle dt = -\phi(0)\langle \zeta, y^N(0)\rangle - \int_0^T \dot{\phi}\langle \zeta, y^N\rangle dt.$$

Choosing first $\phi \in C_c^\infty((0,T))$ and exploiting (3.2.8) we have

$$\int_0^T \phi\langle \zeta, \dot{y}\rangle dt = -\int_0^T \dot{\phi}\langle \zeta, y\rangle dt.$$

This means exactly that $y \in H^1((0,T),\mathcal{Z})$ and $\dot{y} = z$. In particular for $\phi \in C_c^\infty([0,T))$ generic we have

$$\int_0^T \phi\langle \zeta, \dot{y}\rangle dt = -\phi(0)\langle \zeta, y(0)\rangle - \int_0^T \dot{\phi}\langle \zeta, y\rangle dt.$$

Then we get also

$$\langle \zeta, y^N(0)\rangle \to \langle \zeta, y(0)\rangle.$$

Since $y^N(0) = 0$ also $y(0) = 0$. □

We now discuss the convergence of p^N.

Lemma 3.18. Up to extracting a further subsequence $u^N \to u$ in $C([0,T], H^1(\Omega)^d)$ and $U^N \to U$ in $C([0,T], L^2(\Omega, \mathrm{DevSym}))$. Moreover $p^N \to p$ in $C([0,T], \mathcal{Q})$ for some p.

Proof. Since $y^N \rightharpoonup y$ in $L^\infty((0,T),\mathcal{Z})$ and $\dot{y}^N \rightharpoonup \dot{y}$ in $L^2((0,T),\mathcal{Z})$ then also

$$\begin{aligned} u^N \rightharpoonup u \text{ in } L^\infty(\Omega, H^2) \quad &\text{and} \quad \dot{u}^N \rightharpoonup \dot{u} \text{ in } L^2(\Omega, H^2),\\ U^N \rightharpoonup U \text{ in } L^\infty(\Omega, H^1) \quad &\text{and} \quad \dot{U}^N \rightharpoonup \dot{U} \text{ in } L^2(\Omega, H^1). \end{aligned}$$

By Aubin–Lions–Simon lemma (Theorem C.2)

$$u^N \to u \text{ in } C([0,T], H^1) \quad \text{and} \quad U^N \to U \text{ in } C([0,T], L^2).$$

We recall that by (3.2.5) we have

$$p^N_n = \frac{1}{d}\operatorname{tr}\left[\tau_N \mathfrak{C}\left(E\dot{u}^N_n - \dot{U}^N_n\right)\right] - \frac{1}{d}\operatorname{tr}\left[\mathfrak{C}\left(Eu^N_n - U^N_n\right)\right]. \tag{3.2.9}$$

Since $\tau_N \to 0$ and $\dot{y}^N_n$ is bounded in $L^2((0,T),\mathcal{Z})$ we have

$$p^N \to p = -\frac{1}{d}\operatorname{tr}\left[\mathfrak{C}\left(Eu - U\right)\right] \quad \text{in } L^2((0,T),\mathcal{Z}).$$

This concludes the proof. □

Remark 3.19. The fact that $p \in C^0([0,T],\mathcal{Q})$ suggest that there are not bifurcation modes of pressure.

Now we notice that (3.2.6) implies that for all $\tilde{z} \in L^2((0,T),\mathcal{Z})$ we have

$$\int_0^T \langle \mathcal{A}\overline{y}^N, \tilde{z}\rangle dt + \int_0^T \langle \mathcal{E}\dot{y}^N, \tilde{z}\rangle dt + \int_0^T \langle \mathcal{D}(P^N, \dot{y}^N), \tilde{z}\rangle dt + \int_0^T \langle \mathcal{B}p^N, \tilde{z}\rangle dt = \int_0^T \langle l^N, \tilde{z}\rangle dt.$$

By Lemma 3.15 and (3.2.8) we have

$$\begin{aligned} \int_0^T \langle l^N, \tilde{z}\rangle dt &\to \int_0^T \langle l, \tilde{z}\rangle dt\\ \int_0^T \langle \mathcal{A}\overline{y}^N, \tilde{z}\rangle dt &\to \int_0^T \langle \mathcal{A}y, \tilde{z}\rangle dt\\ \int_0^T \langle \mathcal{E}\dot{y}^N, \tilde{z}\rangle dt &\to \int_0^T \langle \mathcal{E}\dot{y}, \tilde{z}\rangle dt\\ \int_0^T \langle \mathcal{B}p^N, \tilde{z}\rangle dt &\to \int_0^T \langle \mathcal{B}p, \tilde{z}\rangle dt. \end{aligned}$$

Moreover we have

$$\begin{aligned} \int_0^T \|\mathcal{D}(p^N, \dot{y}^N)\|^2_{\mathcal{Z}'} dt &\leqslant \int_0^T \int_\Omega Y(p^N)^2 |\partial G(\dot{U}^N)|^2 dx dt \leqslant\\ &\leqslant \int_0^T \int_\Omega (Y(0) + \mathrm{Lip}(Y)|p^N|)^2 \|\partial G\|^2_\infty dx dt \lesssim\\ &\lesssim (1 + \|p^N\|^2_{L^2((0,T),\mathcal{Q})}). \end{aligned}$$

But since p^N converges in $C([0,T],\mathcal{Q})$ then $\mathcal{D}(p^N, \dot{y}^N)$ is bounded in $L^2((0,T),\mathcal{Z}')$. Then up to extracting a further subsequence

$$\mathcal{D}(p^N, \dot{y}^N) \rightharpoonup \delta \text{ in } L^2((0,T),\mathcal{Z}')$$

for some $\delta \in L^2((0,T),\mathcal{Z}')$. Then we have

$$\int_0^T \langle \mathcal{A}y,\tilde{z}\rangle dt + \int_0^T \langle \mathcal{E}\dot{y},\tilde{z}\rangle dt + \int_0^T \langle \delta,\tilde{z}\rangle dt + \int_0^T \langle \mathcal{B}p,\tilde{z}\rangle dt = \int_0^T \langle l,\tilde{z}\rangle dt \tag{3.2.10}$$

for all $\tilde{z} \in L^2((0,T),\mathcal{Z})$. This implies also

$$\langle \mathcal{A}y,\tilde{z}\rangle + \langle \mathcal{E}\dot{y},\tilde{z}\rangle + \langle \delta,\tilde{z}\rangle + \langle \mathcal{B}p,\tilde{z}\rangle = \langle l,\tilde{z}\rangle$$

for almost every $t \in (0,T)$. If we prove that $\delta = \mathcal{D}(p,\dot{y})$ we have that (y,p) is a global in time solution to Problem 3.4 and we have concluded. To this aim let us define $\mathcal{Z}_T = L^2((0,T),\mathcal{Z})$ and $\mathcal{Q}_T = L^2((0,T),\mathcal{Q})$. Let $\mathcal{J}\colon \mathcal{Q}\times\mathcal{Z} \to \mathbb{R}$ be defined by

$$\mathcal{J}(p,\dot{y}) = \int_\Omega Y(p)G(\dot{U})dx.$$

Let also $\mathcal{J}_T\colon \mathcal{Q}_T\times\mathcal{Z}_T \to \mathbb{R}$ and $\mathcal{D}_T\colon \mathcal{Q}_T\times\mathcal{Z}_T \to \mathcal{Z}'_T$ defined by

$$\mathcal{J}_T(p,\dot{y}) = \int_0^T \mathcal{J}(p,\dot{y})dt \quad \text{and} \quad \langle \mathcal{D}_T(p,\dot{y}),\tilde{z}\rangle = \int_0^T \langle \mathcal{D}(p,\dot{y}),\tilde{z}\rangle dt$$

for all $\tilde{z} \in L^2((0,T),\mathcal{Z})$. We notice that for all $y \in \mathcal{Z}_T$ and $p \in \mathcal{Q}_T$ we can identify $\mathcal{D}_T(p,\dot{y}) \in \mathcal{Z}'_T$ with $\mathcal{D}(p,\dot{y}) \in L^2((0,T),\mathcal{Z}')$.

Lemma 3.20. Let $\dot{y} \in L^2((0,T),\mathcal{Z})$, $p \in L^2((0,T),\mathcal{Q})$ and $\delta \in L^2((0,T),\mathcal{Z}')$. Then $\delta = \mathcal{D}(p,\dot{y})$ if and only if

$$\mathcal{J}_T(p,\tilde{z}) \geqslant \mathcal{J}_T(p,\dot{y})dt + \int_0^T \langle \delta,\tilde{z}-\dot{y}\rangle dt \tag{3.2.11}$$

for all $\tilde{z} \in L^2((0,T),\mathcal{Z})$.

Proof. It is easy to see that $\mathcal{J}$ is convex in $\dot{y}$ and is Gateaux differentiable in $\dot{y}$ with Gateaux differential

$$\langle D_{\dot{y}}\mathcal{J}_T(p,\dot{y}),\tilde{z}\rangle = \int_0^T \langle \mathcal{D}(p,\dot{y}),\tilde{z}\rangle dt = \langle \mathcal{D}_T(p,\dot{y}),\tilde{z}\rangle$$

for all $\tilde{z} \in L^2((0,T),\mathcal{Z})$. By [13, Proposition 5.3] then $\mathcal{J}_T$ is sub differentiable and $D^{\mathrm{sub}}_{\dot{y}}\mathcal{J}_T(p,\dot{y}) = \{D_{\dot{y}}\mathcal{J}_T(p,\dot{y})\}$. Then $\delta = \mathcal{D}(p,\dot{y})$ if and only if $\delta \in D^{\mathrm{sub}}_{\dot{y}}\mathcal{J}_T(p,\dot{y})$ but this last inclusion is equivalent to (3.2.11). ◻

By that lemma we have

$$\mathcal{J}_T(p^N,\tilde{z}) \geqslant \mathcal{J}_T(p^N,\dot{y}) + \langle \mathcal{D}_T(p^N,\dot{y}^N),\tilde{z}-\dot{y}^N\rangle. \tag{3.2.12}$$

Now we want to pass to pass to the limit in this inequality. We will need the following lemma.

Lemma 3.21. For each $\dot{y} \in \mathcal{Z}_T$ the functional $\mathcal{J}_T(\cdot,\dot{y})$ is Lipschitz continuous with Lipschitz constant bounded by $\|\dot{y}\|_{\mathcal{Z}_T}$. Moreover for each $p \in \mathcal{Q}_T$ we have that $\mathcal{J}_T(p,\cdot)$ is convex and continuous. In particular $\mathcal{J}_T(p,\cdot)$ is weakly lower semicontinuous.

Proof. We have that

$$\begin{aligned}
|\mathcal{J}_T(p_2,\dot{y}) - \mathcal{J}_T(p_1,\dot{y})| &\leqslant \int_0^T\int_\Omega |Y(p_2)-Y(p_1)||G(\dot{U})|dx \leqslant \\
&\leqslant \|\partial G\|_\infty \mathrm{Lip}(Y)\int_0^T\int_\Omega |p_2-p_1||\dot{U}|dtdx \lesssim \\
&\lesssim \|p_2-p_1\|_{\mathcal{Q}_T}\|\dot{y}\|_{\mathcal{Z}_T}.
\end{aligned}$$

This proves that for each $\dot{y} \in \mathcal{Z}_T$ the functional $\mathcal{J}_T(\cdot, \dot{y})$ is Lipschitz continuous with Lipschitz constant bounded by $\|\dot{y}\|_{\mathcal{Z}_T}$. The convexity of $\mathcal{J}_T(p, \cdot)$ follows immediately from the convexity of G. Moreover

$$
\begin{aligned}
|\mathcal{J}_T(p, \dot{y}_2) - \mathcal{J}_T(p, \dot{y}_1)| &\leqslant \int_0^T \int_\Omega |Y(p)||G(\dot{U}_2) - G(\dot{U}_1)|\,dx \leqslant \\
&\leqslant \|\partial G\|_\infty \int_0^T \int_\Omega (Y(0) + \mathrm{Lip}(Y)|p|)|\dot{U}_2 - \dot{U}_1|\,dt\,dx \lesssim \\
&\lesssim (1 + \|p\|_{\mathcal{Q}_T})\|\dot{y}_2 - \dot{y}_1\|_{\mathcal{Z}_T}.
\end{aligned}
$$

This proves the continuity of $\mathcal{J}_T(p, \cdot)$ for each $p \in \mathcal{Q}_T$ fixed. □

Now we have
$$|\mathcal{J}_T(p^N, \tilde{z}) - \mathcal{J}_T(p, \tilde{z})| \leqslant C\|p^N - p\|_{\mathcal{Q}_T} \to 0$$
for some constant $C > 0$. Moreover

$$\mathcal{J}_T(p^N, \dot{y}^N) = [\mathcal{J}_T(p^N, \dot{y}^N) - \mathcal{J}_T(p, \dot{y}^N)] + \mathcal{J}_T(p, \dot{y}^N).$$

But since $p^N \top$ in $\mathcal{Q}_T$ and y^N are bounded in $\mathcal{Z}_T$

$$|\mathcal{J}_T(p^N, \dot{y}^N) - \mathcal{J}_T(p, \dot{y}^N)| \leqslant C\|y^N\|_{\mathcal{Z}_T}\|p^N - p\|_{\mathcal{Q}_T} \to 0$$

and
$$\liminf_{N\to\infty} \mathcal{J}_T(p, \dot{y}^N) \geqslant \mathcal{J}_T(p, \dot{y}).$$

Then
$$\liminf_{N\to\infty} \mathcal{J}_T(p^N, \dot{y}^N) \geqslant \mathcal{J}_T(p, \dot{y}).$$

Now by construction

$$
\begin{aligned}
\int_0^T \langle \mathcal{A}y^N, \tilde{z} - \dot{y}^N\rangle dt + \int_0^T \langle \mathcal{E}\dot{y}^N, \tilde{z} - \dot{y}^N\rangle dt + \int_0^T \langle \mathcal{D}(p^N, \dot{y}^N), \tilde{z} - \dot{y}^N\rangle dt + \\
+ \int_0^T \langle \mathcal{B}p^N, \tilde{z} - \dot{y}^N\rangle dt = \int_0^T \langle l^N, \tilde{z} - \dot{y}^N\rangle dt.
\end{aligned}
$$

But by Lemma 3.15, Lemma 3.18 and (3.2.8) we have

$$
\begin{aligned}
\int_0^T \langle \mathcal{A}y^N, \tilde{z} - \dot{y}^N\rangle dt &\to \int_0^T \langle \mathcal{A}y, \tilde{z} - \dot{y}\rangle dt \\
\int_0^T \langle \mathcal{B}p^N, \tilde{z} - \dot{y}^N\rangle dt &\to \int_0^T \langle \mathcal{B}p, \tilde{z} - \dot{y}\rangle dt \\
\int_0^T \langle l^N, \tilde{z} - \dot{y}^N\rangle dt &\to \int_0^T \langle l, \tilde{z} - \dot{y}\rangle dt
\end{aligned}
$$

since the convergence of y^N, p^N and l^N is strong. Moreover since

$$\dot{y} \to \int_0^T \langle \mathcal{E}\dot{y}^N, \dot{y}^N - \tilde{z}\rangle dt$$

is continuous and convex in $L^2((0,T), \mathcal{Z})$ it is weakly lower semicontinuous and

$$\liminf_{N\to\infty} \int_0^T \langle \mathcal{E}\dot{y}^N, \dot{y}^N - \tilde{z}\rangle dt \geqslant \int_0^T \langle \mathcal{E}\dot{y}, \dot{y} - \tilde{z}\rangle dt.$$

This implies that

$$\liminf_{N\to\infty}\langle \mathcal{D}_T(p^N, \dot{y}^N), \tilde{z}-\dot{y}^N\rangle \geqslant$$
$$\geqslant \int_0^T \langle l, \tilde{z}-\dot{y}\rangle dt - \int_0^T \langle \mathcal{A}y, \tilde{z}-\dot{y}\rangle dt - \int_0^T \langle \mathcal{E}\dot{y}, \tilde{z}-\dot{y}\rangle dt - \int_0^T \langle \mathcal{B}p, \tilde{z}-\dot{y}\rangle dt =$$
$$= \int_0^T \langle \delta, \tilde{z}-\dot{y}\rangle dt$$

where the last equality holds for (3.2.10). Collecting these results when we take the inferior limit for $N \to \infty$ of (3.2.12) and we get

$$\mathcal{J}_T(p, \tilde{z}) \geqslant \mathcal{J}_T(p, \dot{y}) + \int_0^T \langle \delta, \tilde{z}-\dot{y}\rangle dt.$$

Since now δ satisfy (3.2.11) we have by Lemma 3.2.11 that $\delta = \mathcal{D}(p, \dot{y})$ as desired.

3.2.5 Example of constitutive relations

In this section we give examples of constitutive relations satisfying the hypotheses of Section 3.2.1. Let $g\colon [0,\infty) \to \mathbb{R}$ be defined by

$$g(r) = \begin{cases} \frac{r^2}{4c} & \text{if } r \leqslant c \\ r - \frac{3}{4}c - \frac{c}{2}\log\frac{r}{c} & \text{if } r > c \end{cases}$$

where $c > 0$ so that

$$g'(r) = \begin{cases} \frac{r}{2c} & \text{if } r \leqslant c \\ 1 - \frac{c}{2r} & \text{if } r > c. \end{cases}$$

We set $G(\dot{U}) = g(|\dot{U}|)$. We let the yield strength be given by

$$Y(p) = \begin{cases} Y_0 & \text{if } p < -p_0 \\ Y_0 + \alpha(p + p_0) & \text{if } p \geqslant -p_0 \end{cases}$$

where $Y_0 \geqslant 0$, $p_0 \geqslant 0$ and $\alpha > 0$. We set $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{D}$, $\mathfrak{E}$ and $\mathfrak{F}$ positive constants and $\mathfrak{C}$ and $\mathfrak{D}$ positive definite isotropic fourth order tensors.

3.2.6 Further regularity hypotheses

The local existence and uniqueness theorem is obtained under the following additional hypotheses.

(Y′) $Y \in C^{1,1}(\mathbb{R})$

(G′) $G \in C^{2,1}(\mathrm{DevSym})$

3.2.7 Local existence and uniqueness and global uniqueness

Proposition 3.22. The map $\mathcal{F}$ is of class C^1 and the Fréchet derivative is given by

$$\begin{aligned}
\langle D_{\dot y}\mathcal{F}(\dot y, p, y, t)z, (\tilde z, \tilde q)\rangle &= \int_\Omega \mathfrak{D}Ev : E\tilde v dx + \int_\Omega \mathfrak{E}\operatorname{grad}^2 v : \operatorname{grad}^2 \tilde v dx + \\
&\quad + \int_\Omega \mathfrak{F}\mathbb{V} \circ \tilde{\mathbb{V}} dx + \int_\Omega Y(p)\partial^2 G(\dot U)V : \tilde V dx \\
\langle D_p\mathcal{F}(\dot y, p, y, q)q, (\tilde z, \tilde q)\rangle &= \int_\Omega Y'(p)q\partial G(\dot U) : \tilde V dx - \int_\Omega q\tilde\mu dx \\
\langle D_y\mathcal{F}(\dot y, p, y, t)z, (\tilde z, \tilde q)\rangle &= \int_\Omega \mathfrak{C}\left(Ev - V - \frac{\mu}{d}I\right) : \left(E\tilde v - \tilde V - \frac{\tilde\mu}{d}I\right) dx + \\
&\quad + \int_\Omega \mathfrak{A}\operatorname{grad}^2 v \,\vdots\, \operatorname{grad}^2 \tilde v dx + \int_\Omega \mathfrak{B}\mathbb{V} \circ \tilde{\mathbb{V}} dx \\
\langle D_t\mathcal{F}(\dot y, p, y, t), (\tilde z, \tilde q)\rangle &= -\int_\Omega \dot f \cdot \tilde v dx - \int_\Omega \dot h \cdot \tilde v dx.
\end{aligned}$$

Proof. I still need to report in LaTeXthe boring check but it should be true. First we prove the Gateaux differentiability everywhere that ensure also the continuity. Then we prove that the Gateaux differential is continuous. To this aim we use Lemma C.9, the following lemma and the embedding of H^1 in L^6 for $d = 2$ or $d = 3$.

Lemma 3.23. Let $\Omega \subset \mathbb{R}^d$ be open and bounded. Let $\phi \colon \mathbb{R}^m \to \mathbb{R}^l$ be Lipschitz continuous and bounded. Let $f_n \to f$ in $L^p(\Omega, \mathbb{R}^m)$ for some $p \in [1, \infty)$. Then $\phi(f_n) \to \phi(f)$ in $L^q(\Omega, \mathbb{R}^l)$ for all $q \in [p, \infty)$.

Proof. Let $A_n = \{x \in \Omega \mid |f_n(x) - f(x)| < 1\}$. Since convergence in L^p implies convergence in measure then $|\Omega \setminus A_n| \to 0$. Now let $q \in [p, \infty)$. Then if we call $L > 0$ the Lipschitz constant of ϕ and $C = \sup_{\mathbb{R}^m} |\phi|$ we have

$$\begin{aligned}
\|\phi(f_n) - \phi(f)\|_q^q &= \int_{A_n} |\phi(f_n) - \phi(f)|^q dx + \int_{\Omega\setminus A_n} |\phi(f_n) - \phi(f)|^q dx \leqslant \\
&\leqslant \int_{A_n} L|f_n - f|^q dx + 2C|\Omega \setminus A_n|.
\end{aligned}$$

But on A_n we have $|f_n - f|^q \leqslant |f_n - f|^p$ so that

$$\|\phi(f_n) - \phi(f)\|_q^q \leqslant \int_{A_n} L|f_n - f|^p q dx + 2C|\Omega \setminus A_n| \leqslant L\|f_n - f\|_p^p + 2C|\Omega \setminus A_n|.$$

This proves that $\phi(f_n) \to \phi(f)$ in $L^q(\Omega, \mathbb{R}^l)$. □

□

We notice that $\mathcal{F}$ does not depend on λ. We call $\xi = (\dot u, \dot U, \lambda, p)$, $\zeta = (v, V, \mu, q) = (z, q)$ and $\tilde\zeta = (\tilde v, \tilde V, \tilde\mu, \tilde q) = (\tilde z, \tilde q)$ all belonging to $\mathcal{Z} \times \mathcal{Q}$.

Proposition 3.24. For any $\dot y \in \mathcal{Z}$, $p \in \mathcal{Q}$, $y \in \mathcal{Z}$ and $t \in [0, T]$ we have $D_\xi\mathcal{F}(\dot y, p, y, t) \in \operatorname{Iso}(\mathcal{Z} \times \mathcal{Q}, \mathcal{Z}' \times \mathcal{Q}')$.

Proof. We define the bilinear form $\mathfrak{a}\colon \mathcal{Z}\times\mathcal{Z}\to\mathbb{R}$ by

$$\mathfrak{a}(z,\tilde z) = \int_\Omega \mathfrak{D}Ev : E\tilde v dx - \int_\Omega \mathfrak{C}\frac{\mu}{d}I : \left(E\tilde v - \tilde V - \frac{\tilde\mu}{d}I\right) dx + \int_\Omega \mathfrak{E}\,\mathrm{grad}^2 v \,\vdots\, \mathrm{grad}^2 \tilde v dx + \\ + \int_\Omega \mathfrak{F}\mathbb{V}\circ\tilde{\mathbb{V}} dx + \int_\Omega Y(p)\partial^2 G(\dot U)V : \tilde V dx.$$

We define also the bilinear forms $\mathfrak{b}\colon \mathcal{Q}\times\mathcal{Z}\to\mathbb{R}$ and $\mathfrak{c}\colon \mathcal{Q}\times\mathcal{Z}\to\mathbb{R}$ by

$$\mathfrak{b}(p,\tilde z) = \int_\Omega (Y'(p)\partial G(\dot U) : \tilde V - \tilde\mu) q\, dx$$

and

$$\mathfrak{c}(\tilde q, z) = -\int_\Omega \mu\tilde q\, dx = 0.$$

To prove that $D_\xi\mathcal{F}(\dot y, p, y, t) \in \mathrm{Iso}(\mathcal{Z}\times\mathcal{Q}, \mathcal{Z}'\times\mathcal{Q}')$ we have to prove that the mixed elliptic problem

$$\begin{aligned} \mathfrak{a}(z,\tilde z) + \mathfrak{b}(p,\tilde z) &= \langle \mathfrak{l}, \tilde z\rangle \quad \text{for all } \tilde z\in\mathcal{Z} \\ \mathfrak{c}(\tilde q, z) &= \langle \mathfrak{m}, \tilde q\rangle \quad \text{for all } \tilde q \in \mathcal{Q} \end{aligned}$$

is well posed for all $\mathfrak{l}\in\mathcal{Z}'$ and $\mathfrak{m}\in\mathcal{Q}'$. To this aim it is sufficient to check the hypotheses of Theorem C.7. We first check the hypotheses on $\mathfrak{b}$. We have

$$\tilde{\mathcal{N}} = \{\tilde z\in\mathcal{Z} \mid \mathfrak{b}(p,\tilde z) = 0 \text{ for all } p\in\mathcal{Q}\} = \left\{\tilde z\in\mathcal{Z} \;\middle|\; \tilde\mu = Y'(p)\partial G(\dot U) : \tilde V\right\}.$$

Notice that $|Y'(p)\partial G(\dot U) : \tilde V| \leqslant \mathrm{Lip}(Y)\|\partial G\|_\infty |\tilde V| \in L^2$. Now

$$\sup_{q\in\mathcal{Q}\setminus\{0\}} \frac{\int_\Omega (Y'(p)\partial G(\dot U) : \tilde V - \tilde\mu) q\, dx}{\|q\|_Q} = \|Y'(p)\partial G(\dot U) : \tilde V - \tilde\mu\|_2.$$

On the other hand since $\tilde z_0 = (\tilde v_0, \tilde V_0, \tilde\mu_0) \in \tilde{\mathcal{N}}$ if and only if $\tilde\mu_0 = Y'(p)\partial G(\dot U) : \tilde V_0$ we have

$$\begin{aligned} \|\tilde z\|_{\mathcal{Z}/\tilde{\mathcal{N}}} &= \inf_{\tilde z_0\in\tilde{\mathcal{N}}} \left[\|\tilde v + \tilde v_0\|_{1,2} + \|\tilde V + \tilde V_0\|_{1,2} + \|\tilde\mu + Y'(p)\partial G(\dot U) : \tilde V_0\|_2\right] = \\ &= \inf_{\tilde V_0\in H^1} \left[\|\tilde V + \tilde V_0\|_{1,2} + \|\tilde\mu + Y'(p)\partial G(\dot U) : \tilde V_0\|_2\right] \leqslant \\ &\leqslant \|\tilde\mu - Y'(p)\partial G(\dot U) : \tilde V\|_2 \end{aligned}$$

where the last inequality holds since we can choose $\tilde V_0 = -\tilde V$. Then we have

$$\sup_{q\in\mathcal{Q}\setminus\{0\}} \frac{\int_\Omega (Y'(p)\partial G(\dot U) : \tilde V - \tilde\mu) q\, dx}{\|q\|_Q} \geqslant \|\tilde z\|_{\mathcal{Z}/\tilde{\mathcal{N}}}.$$

Moreover if $q\in\mathcal{Q}$ is such that

$$\mathfrak{b}(p,\tilde z) = \int_\Omega (Y'(p)\partial G(\dot U)\circ\tilde V - \tilde\mu) q\, dx = 0 \quad \text{for all } \tilde z\in\mathcal{Z}$$

then choosing $\tilde V = 0$ and $\tilde\mu\in L^2(\Omega)$ arbitrary we have $q = 0$. This completes the check of the hypotheses on $\mathfrak{b}$. The check of the hypotheses on $\mathfrak{c}$ is similar. In particular we have

$$\mathcal{N} = \{z\in\mathcal{Z} \mid \mathfrak{c}(\tilde q, z) = 0 \text{ for all } \tilde q\in\mathcal{Q}\} = \{z\in\mathcal{Z} \mid \mu = 0\}.$$

Now we check the hypotheses on a. For any $z = (v, V, 0) \in \mathcal{N}$ we consider $\tilde{z} = (v, V, Y'(p)\partial G(\dot{U}) : V) \in \tilde{\mathcal{N}}$ and conversely. We notice that with that choice of z and $\tilde{z}$ there exist $K > 0$ and $C > 0$ such that $\|\tilde{z}\|_{\mathcal{Z}} \leqslant K\|z\|_{\mathcal{Z}}$ and

$$a(z, \tilde{z}) \geqslant \delta\|Ev\|_2^2 + \epsilon\|\text{grad}^2 v\|_2^2 + \phi\|\mathbb{V}\|_2^2 \geqslant C\|z\|_{\mathcal{Z}}^2 \geqslant \frac{C}{K^2}\|\tilde{z}\|_{\mathcal{Z}}^2.$$

This immediately implies that for all $z \in \mathcal{N}$ we have

$$\sup_{\tilde{z}\in\mathcal{Z}} \frac{a(z,\tilde{z})}{\|\tilde{z}\|_{\mathcal{Z}}} \geqslant \frac{C}{K}\|z\|_{\mathcal{Z}}$$

and if $\tilde{z} \in \tilde{\mathcal{N}}$ satisfy $a(\cdot, \tilde{z}) = 0$ then $\tilde{z} = 0$. Then also the hypotheses on a are verified. This completes the proof. □

By Proposition 3.24 in a neighborhood of every $(\dot{y}_0, p_0, y_0, t_0)$ with $\lambda_0 = 0$ we can explicit $\xi = (\dot{u}, \dot{U}, \lambda, p)$ from the equation

$$\mathcal{F}(\dot{y}, p, y, t) = 0.$$

We get

$$(\dot{u}, \dot{U}) = \mathcal{G}(u, U, t) \quad \text{and} \quad (\lambda, p) = \mathcal{H}(u, U, t)$$

where $\mathcal{G}$ and $\mathcal{H}$ are maps of class C^1. Then we can apply Theorem C.12 to get locally in time a unique solution (u, U) of class C^2 to the Cauchy problem

$$\begin{cases} (\dot{u}, U) = \mathcal{G}(u, U, t) \\ u(0) = u_0 \\ U(0) = U_0. \end{cases}$$

Then we set $(\lambda, p) = \mathcal{H}(u, U, t)$ which is of class C^1. It is clear that $\mathcal{F}(\dot{y}, p, y, t) = 0$. In conclusion we have obtained the following theorem.

Theorem 3.25. Let $(\dot{y}_0, p_0, y_0, t_0) \in \mathcal{Z} \times \mathcal{Q} \times \mathcal{Z} \times [0, T]$ with $\lambda_0 = 0$. Then there exists a unique C^1 solution (y, p) to Problem 3.4 defined locally around t_0.

Considering also Theorem 3.5 we obtain global uniqueness.

Corollary 3.26. Assume the hypotheses of Section 3.2.1 and of Section 3.2.6. Then there exists a unique $y \in C^1([0, T], \mathcal{Z})$ and $p \in C^1([0, T], \mathcal{Q})$ solution to Problem 3.4.

3.2.8 Bifurcation analysis

Remark 3.27. This section is not definitive but contains only the ideas that we have collected in the hope of being able to reach a more complete result regarding the phenomenon of localization of deformation.

Some considerations

Since we have proved in Corollary 3.26 that the solution is unique no bifurcation can occur with nonzero viscosity. As the viscosity approaches to zero we expect that the evolution of the system reveals it extreme sensibility on the problem data when strain localization occur. To explain better the ideas we present the following example that describe what is the behavior that we expect. Consider a perfectly symmetrical specimen under perfectly symmetrical compressive loads. By uniqueness and symmetry in presence of viscosity the specimen will experience a perfectly symmetrical deformation pattern during the entire evolution (a distributed deformation or at most a symmetrical pattern of localized shear bands). However if the loads, the geometry or the material properties of the specimen are not perfectly symmetric then the system experience an unsymmetrical and completely different deformation pattern (for example an unsymmetrical pattern of shear bands). When viscosity decreases, the threshold beyond which perturbations from the condition of perfect symmetry trigger a localization with a significantly altered pattern is reduced. In some sense in the case of null viscosity the system is sensible also to vanishing perturbations. This suggest a lost of uniqueness at a stage of the loading program also in case of perfect symmetry.

We also expect that the bifurcation is actually associated with a jump in the time evolution. The reasons are the following.

(i) When passing to the limit $N \to \infty$ in the Rothe method in Section 3.2.4 we used the viscosity to get the a priori estimates on $\dot{y}^N$. We where not able to get them without viscosity, using only the $H^1((0,T))$ regularity of the load, as can be done for Von Mises plasticity [22]. This suggest that without viscosity the solution of our model cannot not lies in $H^1((0,T))$ but must lie in $BV((0,T))$.

(ii) Similar models of materials with pressure dependent yield strength exhibit jumps. It is the case of Cam–Clay model studied in [12] or the model studied analytically in [11] coupling the associative Drucker–Prager model with damage.

(iii) We expect that when the viscosity is small the evolution became very fast where the deformation occur. In the limit of null viscosity this became a jump.

Remark 3.28. My desire is to understand better the phenomenon of localization when the viscosity if small but nonzero trying to understand how the sensitivity of the system to imperfection increases with decreasing viscosity. Anyway, if it is useful in some way, I will try to study also the case of zero viscosity. But I would like to avoid the use of the notion of energetic solution. Indeed the choice made by the system at the jump point in the energetic solution approach is determined by a global minimizing approach that seems nonphysical to us. This view is confirmed in [25]. The approach that I could accept is that of balanced viscosity solutions where is the viscosity that drive the evolution at the jumps.

Study on the invertibility of the stiffness operator

In what follows are reported our efforts to understand better the phenomenon of localization. In particular we have confined ourself to the time discrete problem (based on the implicit Euler scheme). We computed the "stiffness operator" (the one that would be assembled in the numerical approximation of the model with Newton method). It seems to became singular during plastic flow in absence of viscosity suggesting that the system reaches indeed a bifurcation point (where the jump occur). This analysis should be intended only as a first trail and not as a final result. We hope that the analysis can be extended also to the the time continuous case.

We consider $\mathfrak{D}_\eta$, $\mathfrak{E}_\eta$ and $\mathfrak{F}_\eta$ as dependent continuously a viscosity parameter $\eta > 0$ and vanishing for $\eta = 0$. We call $\mathcal{F}_\eta$ the corresponding functional. We consider the discrete in time problem (Problem 3.6). We focus on a single time step and to simplify the notation we set $y = y^N_{n-1}$, $t = t^N_{n-1}$, $p = p^N_n$ and $\dot{y} = \tau_N^{-1}(y^N_n - y^N_{n-1})$ and $\tau = \tau_N$. Then $y^N_n = y + \tau\dot{y}$. The incremental problem reduces to find $\dot{y} \in \mathcal{Z}$ and $p \in \mathcal{Q}$ such that

$$\mathcal{F}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p) = \mathcal{F}_\eta(\dot{y}, p, y + \tau\dot{y}, t + \tau) = 0.$$

We compute the Fréchet differential $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y},p)$ of $\mathcal{F}^{(y,t)}_{(\tau,\eta)}$ with respect to $(\dot{y},p)$. It can be written as

$$\langle \mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y},p)(z,q),(\tilde{z},\tilde{q})\rangle = a^{(\dot{y},p,y,t)}_{(\tau,\eta)}(z,\tilde{z}) + b^{(\dot{y},p,y,t)}_{(\tau,\eta)}(q,\tilde{z}) + c^{(\dot{y},p,y,t)}_{(\tau,\eta)}(\tilde{q},z).$$

We have introduced the bilinear form

$$\begin{aligned} a^{(\dot{y},p,y,t)}_{(\tau,\eta)}(z,\tilde{z}) = &\int_\Omega \mathfrak{D}_\eta Ev : E\tilde{v}dx + \int_\Omega \tau\mathfrak{C}\left(Ev - V - \frac{\mu}{d}I\right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I\right)dx + \\ &+ \int_\Omega (\mathfrak{E}_\eta + \tau\mathfrak{A})\,\mathrm{grad}^2 v : \mathrm{grad}^2\tilde{v}dx + \int_\Omega (\mathfrak{F}_\eta + \tau\mathfrak{B})\mathbb{V}\circ\tilde{\mathbb{V}}dx + \\ &+ \int_\Omega Y(p)\partial^2 G(\dot{U})V : \tilde{V}dx \\ b^{(\dot{y},p,y,t)}_{(\tau,\eta)}(q,\tilde{z}) = &\int_\Omega q(Y'(p)\partial G(\dot{U}) : \tilde{V} - \tilde{\mu})dx \\ c^{(\dot{y},p,y,t)}_{(\tau,\eta)}(q,\tilde{z}) = &-\int_\Omega \tau\mu\tilde{q}dx. \end{aligned}$$

We want to study if $\mathcal{K}^{(y,t)}_{(\tau,\eta)}$ is in $\mathrm{Iso}(\mathcal{Z}\times\mathcal{Q},\mathcal{Z}'\times\mathcal{Q}')$. To this aim we can apply Theorem C.7 to the bilinear forms $a^{(\dot{y},p,y,t)}_{(\tau,\eta)}$, $b^{(\dot{y},p,y,t)}_{(\tau,\eta)}$ and $c^{(\dot{y},p,y,t)}_{(\tau,\eta)}$. Reasoning as in the proof of Proposition 3.24 we have that $b^{(\dot{y},p,y,t)}_{(\tau,\eta)}$ and $c^{(\dot{y},p,y,t)}_{(\tau,\eta)}$ satisfies the hypotheses. Moreover the sets $\mathcal{N}$ and $\tilde{\mathcal{N}}$ are defined exactly in the same way. We need to check the hypotheses on $a^{(\dot{y},p,y,t)}_{(\tau,\eta)}$. We reason again as in the proof of Proposition 3.24. For any $z = (v,V,0) \in \mathcal{N}$ we consider $\tilde{z} = (v,V,Y'(p)\partial G(\dot{U}) : V) \in \tilde{\mathcal{N}}$ and conversely. We have that with that choice of z and $\tilde{z}$ there exist $K > 0$ such that $\|\tilde{z}\|_{\mathcal{Z}} \leqslant K\|z\|_{\mathcal{Z}}$. Then

$$\begin{aligned} a^{(\dot{y},p,y,t)}_{(\tau,\eta)}(z,\tilde{z}) = &\int_\Omega \mathfrak{D}_\eta Ev : Evdx + \int_\Omega \tau\mathfrak{C}(Ev - V) : \left(Ev - V - \frac{Y'(p)\partial G(\dot{U}) : V}{d}I\right)dx + \\ &+ \int_\Omega (\mathfrak{E}_\eta + \tau\mathfrak{A})\,\mathrm{grad}^2 v : \mathrm{grad}^2 vdx + \int_\Omega (\mathfrak{F}_\eta + \tau\mathfrak{B})\mathbb{V}\circ\mathbb{V}dx + \\ &+ \int_\Omega Y(p)\partial^2 G(\dot{U})V : Vdx \\ \leqslant\ &\delta_\eta\|Ev\|_2^2 + \tau\gamma\|Ev - V\|_2^2 - 2\tau\kappa\|Ev - V\|_2\|V\|_2 + \\ &+ (\epsilon_\eta + \tau\alpha)\|\mathrm{grad}^2 v\|_2^2 + (\phi_\eta + \tau\beta)\|\mathbb{V}\|_2^2 + \sigma(p,\dot{U})\|V\|_2^2. \end{aligned}$$

Here

$$\kappa = \frac{|\mathfrak{C}|\mathrm{Lip}(Y)\|\partial G\|_\infty}{2\sqrt{d}} \quad \text{and} \quad \sigma(p,\dot{U}) = \mathrm{essinf}_\Omega\left[Y(p)\ \mathrm{se}(\partial^2 G(\dot{U}))\right] \geqslant 0$$

where se denotes the smallest eigenvalue. Then by Young inequality for all $\theta_1 \in (0,\kappa^{-1}\gamma)$ and $\theta_2 \in (0,\infty)$ we have

$$\begin{aligned} a^{(\dot{y},p,y,t)}_{(\tau,\eta)}(z,\tilde{z}) \leqslant\ &\delta_\eta\|Ev\|_2^2 + \tau(\gamma - \kappa\theta_1)\|Ev - V\|_2^2 - \tau\frac{\kappa}{\theta_1}\|V\|_2^2 + \\ &+ (\epsilon_\eta + \tau\alpha)\|\mathrm{grad}^2 v\|_2^2 + (\phi_\eta + \tau\beta)\|\mathbb{V}\|_2^2 + \sigma(p,\dot{U})\|V\|_2^2. \\ \leqslant\ &\left[\delta_\eta + \tau(\gamma - \kappa\theta_1)(1 - \theta_2)\right]\|Ev\|_2^2 + (\epsilon_\eta + \tau\alpha)\|\mathrm{grad}^2 v\|_2^2 + \\ &+ \left\{\phi_\eta + \sigma(p,\dot{U}) + \tau\left[(\gamma - \kappa\theta_1)\left(1 - \frac{1}{\theta_2}\right) - \frac{\kappa}{\theta_1} + \beta\right]\right\}\|V\|_2^2 + \\ &+ (\phi_\eta + \tau\beta)l^2\|\mathrm{grad}\,V\|_2^2. \end{aligned}$$

The hypotheses on $a^{(\dot{y},p,y,t)}_{(\tau,\eta)}$ are satisfied if we can find $\theta = (\theta_1, \theta_2) \in (0, \kappa^{-1}\gamma) \times (0, \infty)$ such that

$$\begin{cases} \delta_\eta + \tau(\gamma - \kappa\theta_1)(1 - \theta_2) > 0 \\ \phi_\eta + \sigma(p, \dot{U}) + \tau\left[(\gamma - \kappa\theta_1)\left(1 - \frac{1}{\theta_2}\right) - \frac{\kappa}{\theta_1} + \beta\right] > 0. \end{cases}$$

It $\eta > 0$ then for each θ fixed there exists $\tau > 0$ sufficiently small such that these conditions are satisfied. This connected to the fact that the solution is unique as already established in Section 3.2.7. When $\eta = 0$ then $\delta_\eta = 0$ and $\phi_\eta = 0$. If $\sigma(p, \dot{U}) > 0$ then we can choose $\theta_2 < 1$ so that the first inequality is satisfied and for $\tau > 0$ sufficiently small also the second inequality is satisfied. If instead $\sigma(p, \dot{U}) = 0$ then the inequalities does not depend on τ and can be solved only if $\beta\gamma > \kappa^2$. Indeed from the first we have $\theta_2 < 1$ and from the second we get

$$\frac{1}{\theta_2} < 1 - \frac{\frac{\kappa}{\theta_1} - \beta}{\gamma - \kappa\theta_1}.$$

Then we can find $\theta_2 \in (0, 1)$ if and only if

$$1 - \frac{\frac{\kappa}{\theta_1} - \beta}{\gamma - \kappa\theta_1} > 1 \quad \text{or equivalently} \quad \beta - \frac{\kappa}{\theta_1} > 0.$$

Then we can find $\theta_1 \in (0, 1)$ if and only if $\beta\gamma > \kappa^2$. This means that hardening must be strong enough compared to the sensitivity of the yield strength on pressure (notice that κ is proportional to $\mathrm{Lip}(Y)$).

Remark 3.29. In the case of Von Mises plasticity the stiffness operator is an isomorphism for any positive hardening. This due to the fact that the additional term that comes from the dependence of the yield strength on the plastic pressure is absent.

Intuition about the localization process

We now assume that $\eta = 0$ and that G is a smooth and convex dissipation potential that satisfy $G(\dot{U}) = |\dot{U}| + c$ for all $\dot{U} \in \mathrm{DevSym}$ with $|\dot{U}| > R$ where $c \in \mathbb{R}$ and $R > 0$ are two constants. This gives a plastic force that is independent on the rate and equal to the yield strength when the rate is sufficiently hight. So it is a regularization of the rate independent dissipation potential that preserve the existence of a yield limit. We notice that

$$\partial^2 G(\dot{U}) = \frac{|\dot{U}|^2 I - \dot{U} \otimes \dot{U}}{|\dot{U}|^2} \quad \text{for } |\dot{U}| > R$$

so that $\mathrm{se}(\partial^2 G(\dot{U})) = 0$ there. On the contrary $\mathrm{se}(\partial^2 G(\dot{U}))$ must be very hight for $\dot{U}$ near the origin if G is a very close approximation of the rate independent dissipation potential $|\cdot|$. We now consider a loading process. When the load is still small the plastic force is under the yield limit and $\sigma(p, \dot{U})$ is hight. By the analysis we have just performed this ensure that $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ is invertible and this suggest that the incremental problem is uniquely solved. Moreover we expect that the deformation rate it is not localized in some region but distributed all over the body. As the load increases $\mathrm{se}(\partial^2 G(\dot{U}))$ decreases (uniformly in all the body). Then $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ is invertible only for small time steps τ. Assuming that the hardening is sufficiently small, we can conclude that $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ became singular (or is nonsingular only for very small time increments so that the evolution stagnates). This is the signal that we have reached the bifurcation point. Now the system to restore the invertibility of $\mathcal{K}^{(y,t)}_{(\tau,\eta)}(\dot{y}, p)$ and overcome the singularity point makes $\dot{U}$ small everywhere except that in a small shear band. This cause an elastic unloading elsewhere. Then out of the shear band $\mathrm{se}(\partial^2 G(\dot{U}))$ becomes big again. This makes $a^{(\dot{y},p,y,t)}_{(\tau,\eta)}$ coercive because out of the shear band the plastic strain is held back by the coercivity of $\partial^2 G(\dot{U})$. In the shear band the plastic strain is held back by the surrounding regions through the plastic strain gradient.

3.3 Numerical approximation

Chapter 4

Large deformation model

This chapter is not updated and incomplete. If I have time I will try to get an existence theorem also for the large strain model but I give the priority to understand better the localization of deformation in the small strain model.

4.1 Formulation

4.1.1 Kinematics

Let $\Omega \subset \mathbb{R}^d$ be the reference configuration of the body. The macroscopic deformation is a map $\chi\colon \Omega \to \mathbb{R}^d$ satisfying $\det \operatorname{grad} \chi > 0$. We set $F = \operatorname{grad} \chi$, $E = (F^T F - I)/2$ and $M = \operatorname{grad}^2 \chi$. We subdivide $\partial\Omega$ in two disjoint parts, Γ_D and Γ_N. On Γ_D we impose the kinematical constrain $\chi = \overline{\chi}$. The distortion of the microscopic structure is described by the plastic distortion $F_p\colon \Omega \to \mathrm{Lin}^+$. We regard F_p as the elastically relaxed distortion so we define the elastic distortion as $F_e = FF_p^{-1}$. The volumetric distortion of the microscopic structure is described by $J_p = \det F_p$ while the shear distortion by $U_p = J_p^{-1} F_p$. We impose the kinematical constrain $J_p = 1$ and to this aim we introduce the Lagrange multiplier p with the meaning of a plastic pressure. The kinematical constrain can be written also as

$$\int_\Omega (J_p - 1)\tilde{q}\, dx = 0 \quad \text{for all } \tilde{q}\colon \Omega \to \mathbb{R}.$$

4.1.2 Statics

We denote with $\tilde{v}$ the virtual rate of u.

Let $\Omega \subset \mathbb{R}^3$ be the reference configuration of the body. The deformation of the body is a map $\chi\colon \Omega \to \mathbb{R}^d$. We set $v = \dot{\chi}$ and $F = \operatorname{Grad} \chi$ and $L = \dot{F}F^{-1}$. We subdivide $\partial\Omega$ into disjoint parts Γ_D and Γ_N. The plastic distortion is a map $F_p\colon \Omega \to \mathrm{Lin}^+$. We set $L_p = \dot{F}_p F_p^{-1}$ and we assume that $L_p = D_p \in \mathrm{Sym}$. We set $U_p = \operatorname{dev} D_p$ and $M_p = \operatorname{Grad} U_p \cdot F_p^{-1}$. The elastic distortion is $F_e = FF_p^{-1}$ and we set $L_e = \dot{F}_e F_e^{-1}$.

Lemma 4.1. It holds $L = L_e + F_e D_p F_e^{-1}$.

Proof. We have $L = \dot{F}F^{-1} = (\dot{F}_e F_p + F_e \dot{F}_p) F_p^{-1} F_e^{-1} = \dot{F}_e F_e^{-1} + F_e \dot{F}_p F_p^{-1} F_e^{-1} = L_e + F_e D_p F_e^{-1}$. □

We add the kinematical constrain $\operatorname{tr} D_p = 0$. To impose this constrain we consider a

Lagrange multiplier $p\colon \Omega \to \mathbb{R}$. The constrain can be written in weak form as

$$-\int_\Omega \operatorname{tr} L_p \tilde{q}\, dX = 0 \quad \text{for all } \tilde{q}\colon \Omega \to \mathbb{R}.$$

4.1.3 Statics

We choose an internal virtual power of the form

$$\tilde{\mathcal{W}} = \int_\Omega T : \tilde{L}_e\, dX + \int_\Omega \left(T_p : \tilde{U}_p - p \operatorname{tr} \tilde{L}_p + K_p \,\vdots\, \tilde{M}_p\right) dX.$$

Here T is the macrostress, T_p is the microstress which can be assumed deviatoric and K_p is the microhyperstress. We choose an external virtual power of the form

$$\tilde{\mathcal{P}} = \int_\Omega f \cdot \tilde{v}\, dX + \int_{\Gamma_N} h \cdot \tilde{v}\, dA + \int_{\Gamma_D} r \cdot \tilde{v}\, dA.$$

Definition 4.2 (Principle of Virtual Powers)**.** Equilibrium holds when $\tilde{\mathcal{W}} = \tilde{\mathcal{P}}$ for all virtual rates.

To get the local version of the equilibrium condition we apply Lemma 4.1 and we write the internal virtual power as

$$\tilde{\mathcal{W}} = \int_\Omega TF^{-T} : \operatorname{Grad} \tilde{v}\, dX + \int_\Omega \left(-T_e : \tilde{L}_p + T_p : \tilde{U}_p - p \operatorname{tr} \tilde{L}_p + K_p \,\vdots\, \tilde{M}_p\right) dX$$

where $T_e = F_e^T T F_e^{-T}$ is the Mandel stress. Then we apply the Green theorem

$$\begin{aligned} \tilde{\mathcal{W}} = -\int_\Omega \operatorname{Div}(TF^{-T}) : \operatorname{Grad} \tilde{v}\, dX + \int_{\partial\Omega} (TF^{-T})N \cdot \tilde{v}\, dA + \\ + \int_\Omega \left\{\left[T_p - \operatorname{dev} T_e - \operatorname{Div}\left(K_p \cdot F_p^{-T}\right)\right] : \tilde{U}_p - \left[\frac{1}{d} \operatorname{tr} T_e + p\right] \operatorname{tr} \tilde{L}_p\right\} dX + \\ + \int_{\partial\Omega} \left(K_p \cdot F_p^{-T} N\right) : \tilde{U}_p\, dA. \end{aligned}$$

Then the local form of the equilibrium condition is

$$\begin{aligned} &\operatorname{Div}(TF^{-T}) + f = 0 \quad \text{in } \Omega \\ &(TF^{-T})N = r \quad \text{in } \Gamma_D \\ &(TF^{-T})N = h \quad \text{in } \Gamma_N \\ &T_p - \operatorname{dev} T_e - \operatorname{Div}\left(K_p \cdot F_p^{-T}\right) = 0 \quad \text{in } \Omega \\ &\frac{1}{d} \operatorname{tr} T_e + p = 0 \quad \text{in } \Omega \\ &K_p \cdot F_p^{-T} N = 0 \quad \text{in } \partial\Omega. \end{aligned}$$

Appendix A

Notation

A.1 Tensors

$\mathrm{Lin} = \mathbb{R}^{d \times d}$	$d \times d$ matrices
Lin^+	$d \times d$ matrices with positive determinant
Sym	$d \times d$ symmetric matrices
Dev	$d \times d$ deviatoric matrices
DevSym	$d \times d$ symmetric and deviatoric matrices

Appendix B

Modeling in continuum mechanics

In this chapter we describe the general procedure to develop a thermodynamically consistent mechanical model. The discussion will be purposely informal since the emphasis of this chapter is on modeling.

B.1 General procedure to build thermodynamically consistent mechanical models

We subdivide the discussion in three parts corresponding to the treatment of kinematics, statics and constitutive relations.

Kinematics Consider a general mechanical system. Its kinematical configuration is described by elements $z \in \mathbf{Z}$ of a vector space $\mathbf{Z}$.

Statics The statics concerns with the instantaneous equilibrium between the internal and external action insisting on the configuration of the systems. The internal and external actions on the configuration of the system are described as a linear functionals, $\mathcal{W} \in \mathbf{Z}^*$ and $\mathcal{P} \in \mathbf{Z}^*$ respectively. The equilibrium between internal and external actions is written as $\mathcal{W} = \mathcal{P}$. This can be written as

$$\tilde{\mathcal{W}} \doteq \langle \mathcal{W}, \tilde{w} \rangle = \langle \mathcal{P}, \tilde{w} \rangle \doteq \tilde{\mathcal{P}} \quad \text{for all } \tilde{w} \in \mathbf{Z}$$

which is the classical statement of the principle of virtual powers. We refer to the test vector $\tilde{w} \in \mathbf{Z}$ as the virtual velocity and to $\tilde{\mathcal{W}}$ and $\tilde{\mathcal{P}}$ as the virtual internal end external power respectively.

Constitutive relations While the external actions $\mathcal{P}$ are assigned as a function $\mathcal{P}\colon [0,T] \to \mathbf{Z}^*$ the internal actions are put in relation with the kinematics of the system through constitutive relations. Since we always neglect inertial effects we assume a constitutive relation giving the internal actions as a set valued map $\mathcal{W}\colon \mathbf{Z} \times \mathbf{Z} \rightsquigarrow \mathbf{Z}$ of z and $\dot{z}$. To obtain constitutive relations compatible with thermodynamics we postulate the existence of a

(i) free energy functional $\Psi\colon \mathbf{Z} \to (-\infty, \infty]$ which is differentiable on its domain,

(i) subdifferentiable dissipation potential $\Phi\colon \mathbf{Z} \times \mathbf{Z} \to [0, \infty]$ such that $\Phi(z, \cdot)$ is convex and $\Phi(z, 0) = 0$, and we define

$$\mathcal{W}(z,\dot z) = \partial\Psi(z) + \partial_{\dot z}\Phi(z,\dot z).$$

Proposition B.1. Let $z\colon [0,T] \to \mathbf{Z}$ be a process satisfying the equilibrium condition

$$\partial\Psi(z(t)) + \partial_{\dot z}\Phi(z(t),\dot z(t)) \ni \mathcal{P}(t)$$

for every $t \in [0,T]$. Let the dissipation rate be defined as $\mathcal{D} = \langle \mathcal{W}, \dot z\rangle - \frac{d}{dt}\Psi(z)$. Then $\mathcal{D} \geqslant 0$.

Proof. Let $\xi(t) \in \partial_{\dot z}\Phi(z(t),\dot z(t))$ such that $\partial\Psi(t) + \xi(t) = \mathcal{P}(t)$ for all $t \in [0,T]$. Then

$$\mathcal{D} = \langle \partial\Psi(z), \dot z\rangle + \langle \xi, \dot z\rangle - \frac{d}{dt}\Psi(z) = \langle \xi, \dot z\rangle.$$

But by definition of subdifferential we have $\Phi(z,0) \geqslant \Phi(z,\dot z) - \langle \xi, \dot z\rangle$ so that $\langle \xi, \dot z\rangle \geqslant \Phi(z,\dot z) \geqslant 0$. This completes the thesis. □

Remark B.2. The inequality $\mathcal{D} \geqslant 0$ is known as the dissipation inequality and has the following meaning: the power $\langle \mathcal{W}, \dot z\rangle$ absorbed (or released) by the system is stored reversibly as (or extracted from the) free energy up to a positive irreversible dissipation.

B.2 Treating constrains with the methods of Lagrange multipliers

Linear constraints can be added in the system by substituting a $\mathbf{Z}$ with a subspace. Nonlinear constrains can be handled by choosing appropriately the free energy or the dissipation potential. However sometimes it can be useful to treat the constrain reaction as an additional unknown of the system. This happen especially if it is appropriate to include constitutive relations depending on the reactions.

Kinematics of constrain Let $\mathbf{K}$ be a convex subset of $\mathbf{Z}$. We consider the constrain $z \in \mathbf{K}$.

Constrain reaction In addiction to the internal and external actions, when we treat a constrain with the method of Lagrange multipliers we must add also the constrain reaction $\mathcal{R} \in \mathbf{Z}^*$. Then the equilibrium condition can be written as

$$\mathcal{W} = \mathcal{P} + \mathcal{R}.$$

We make the assumption that the power expanded by the reaction vanish requiring that $\mathcal{R} \in N_{\mathbf{K}}(z)$. The constrain reaction is an additional descriptor of the system together with the kinematical descriptor z.

Constitutive relations In presence of Lagrange multipliers the constitutive relation for $\mathcal{W}$ can be formulated as

$$\mathcal{W}(z,\dot z) = \partial\Psi(z) + \partial_{\dot z}\Phi(\mathcal{R}, z,\dot z).$$

Here the free energy Ψ is exactly as in the unconstrained case and the dissipation potential Φ is exactly as before but include a dependence on $\mathcal{R}$. The dissipation inequality is still valid in the following version.

Proposition B.3. Let $z\colon [0,T] \to \mathbf{Z}$ and $\mathcal{R}\colon [0,T] \to \mathbf{Z}^*$ satisfying

$$z(t) \in \mathbf{K}, \quad \mathcal{R}(t) \in N_{\mathbf{K}}(z(t)) \quad \text{and} \quad \partial\Psi(z(t)) + \partial_{\dot z}\Phi(\mathcal{R}(t), z(t),\dot z(t)) \ni \mathcal{P}(t) + \mathcal{R}(t)$$

for all $t \in [0,T]$. Let the dissipation rate be defined as $\mathcal{D} = \langle \mathcal{W}, \dot z\rangle - \frac{d}{dt}\Psi(z)$. Then $\mathcal{D} \geqslant 0$.

Appendix C

Tools from functional analysis

In this chapter we collect some tools of linear and nonlinear analysis used in the main body of this thesis.

C.1 Linear functional analysis

C.1.1 Other

Proposition C.1. Let $\Omega \subset \mathbb{R}^d$ be open and bounded. Let $p \in [1, \infty]$. Let $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$. Then $u_n \to u$ in $L^p(\Omega)$.

Proof. Since $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$ then u_n are bounded in $W^{1,p}(\Omega)$. By Rellich–Kondrakov theorem each subsequence of u_n admits a further subsequence that converges strongly in $L^p(\Omega)$. Since we already know the weak convergence in $W^{1,p}(\Omega)$ then the limit is always u. Then necessarily the whole sequence u_n converges strongly to u in $L^2(\Omega)$. □

C.1.2 The Aubin–Lions–Simon lemma

Theorem C.2 (Aubin–Lions–Simon lemma)**.** Let $T > 0$ and $V \subset H \subset E$ be Banach spaces. Assume that the embedding of V in H is compact and the embedding of H in E is compact. Let p and q in $[1, \infty]$. Then the space

$$\{u \in L^p((0,T), V) \mid \dot{u} \in L^q((0,T), E)\}$$

i compactly embedded in

(i) $L^p((0,T), H)$ if $p < \infty$

(ii) $C([0,T], H)$ if $p = \infty$ and $q > 1$.

Proof. See [7, Theorem II.5.16]. □

C.1.3 Generalized mixed elliptic problems

We start with the following theorem.

Theorem C.3 (Babuŝka)**.** Let V and Q be reflexive Banach spaces. Let $b\colon Q \times V \to \mathbb{R}$ be bilinear form. Let $Z = \{z \in V | b(q,z) = 0 \text{ for all } q \in Q\}$ and define $W = V/Z$. Assume that b is continuous so that there exists $M > 0$ such that

$$b(q,v) \leqslant M\|q\|_Q\|v\|_V \quad \text{for all } (q,v) \in Q \times V,$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q \in Q \setminus \{0\}} \frac{b(q, w)}{\|q\|_Q} \geqslant \beta \|w\|_W$$

for all $w \in W$ and that if $q \in Q$ satisfies

$$b(q, v) = 0 \quad \text{for all } v \in V$$

then $q = 0$. Then for each $f \in Q'$ there exists a unique $w \in W$ such that

$$b(q, w) = \langle f, q \rangle \quad \text{for all } q \in Q$$

and

$$\|w\|_W \leqslant \frac{1}{\beta} \|f\|_{Q'}.$$

Proof. Step 1. Consider the linear operator $B \colon W \to Q'$ defined by

$$\langle Bw, q \rangle = b(w, q).$$

We notice that

$$|\langle Bw, q \rangle| = \inf_{z \in Z} |b(w + z, q)| \leqslant M \|q\|_Q \inf_{z \in Z} \|w + z\|_V = M \|q\|_Q \|w\|_W$$

so that $B \in \mathcal{L}(W, Q')$.

Step 2. We prove that B is injective. Indeed if $w \in W$ satisfies $Bw = 0$ we have

$$\beta \|w\|_W \leqslant \sup_{q \in Q \setminus \{0\}} \frac{\langle Bw, q \rangle}{\|q\|_Q} = 0$$

so that $w = 0$.

Step 3. We prove that range(B) is closed. Indeed let $f_n = Bw_n$ be such that $f_n \to f$ in Q. Then f_n is a Cauchy sequence. Now

$$\beta \|w_n - w_m\| \leqslant \sup_{q \in Q \setminus \{0\}} \frac{\langle B(w_n - w_m), q \rangle}{\|q\|_Q} \leqslant \|f_n - f_m\|_{Q'}.$$

Then also w_n is a Cauchy sequence in W and $w_n \to w$ in W for some $w \in W$. Now by continuity of B we have $Bw = f$ so that $f \in$ range(B).

Step 4. We prove that range$(B) = Q'$. Indeed let $(B^T) = \{q \in Q | b(q, v) = 0 \text{ for all } v \in V\} = \{0\}$ by hypotheses. On the other hand $(B^T) = \text{range}(B)^\perp$. Since Q is reflexive range$(B) = (\text{range}(B)^\perp)^\perp = Q'$.

Step 5. We have proved that B is a bijection from W to Q'. Then for each $f \in Q'$ there exists a unique $w \in W$ such that $Bw = f$. Moreover

$$\beta \|w\|_W \leqslant \sup_{q \in Q \setminus \{0\}} \frac{\langle Bw, q \rangle}{\|q\|_Q} = \sup_{q \in Q \setminus \{0\}} \frac{\langle f, q \rangle}{\|q\|_Q} \leqslant \|f\|_{Q'}.$$

This completes the proof. □

Lemma C.4. Assume that the hypotheses of Theorem C.3 holds. Then

$$\sup_{w \in W \setminus \{0\}} \frac{b(q,w)}{\|w\|_W} \geqslant \beta \|q\|_Q$$

for all $q \in Q$.

Proof. For each $f \in Q'$ let $w_f \in W$ the solution to

$$b(q,w) = \langle f, q \rangle \quad \text{for all } q \in Q.$$

Notice that $\|f\|_{Q'} \geqslant \beta \|w\|_W$. Now

$$\|q\|_Q = \sup_{f \in Q' \setminus \{0\}} \frac{\langle f, q \rangle}{\|f\|_{Q'}} = \sup_{f \in Q' \setminus \{0\}} \frac{b(q, w_f)}{\|f\|_{Q'}} \leqslant \sup_{f \in Q' \setminus \{0\}} \frac{b(q, w_f)}{\beta \|w\|_W} \leqslant \sup_{w \in W \setminus \{0\}} \frac{b(q,w)}{\beta \|w\|_W}.$$

This completes the proof. □

Then we get immediately the following proposition.

Corollary C.5. Assume the hypotheses of Theorem C.3. Then for each $g \in W'$ there exists a unique $p \in Q$ such that

$$b(p,w) = \langle g, w \rangle \quad \text{for all } w \in W$$

and

$$\|p\|_Q \leqslant \frac{1}{\beta} \|g\|_{W'}.$$

Proof. Apply first Lemma C.4 and then Theorem C.3 with the role of W and Q swapped. □

Remark C.6. Assume that $g \in V'$ is such that $\langle g, z \rangle = 0$ for all $z \in Z$. Then we can define g as an element of W'. Indeed for each $w \in W$ we take a representative $v \in V$ still denoted with w and we define $\langle g, w \rangle = \langle g, v \rangle$. The definition does not depend on the choice of the representative. Moreover

$$\|g\|_W = \sup_{w \in W \setminus \{0\}} \frac{\langle g, w \rangle}{\|w\|_W} = \sup_{v \in V \setminus \{0\}} \frac{\langle g, v \rangle}{\|v\|_V} = \|g\|_{V'}.$$

where we use that since V is reflexive for each $w \in W$ there exists a representative $v \in V$ such that $\|w\|_W = \|v\|_V$ and that for any other representative $u \in V$ we have $\|w\|_W \leqslant \|u\|_V$. In this case the Corollary C.5 can be formulated as follows. There exists a unique $p \in Q$ such that

$$b(p,v) = \langle g, v \rangle \quad \text{for all } v \in V$$

and

$$\|p\|_Q \leqslant \frac{1}{\beta} \|g\|_{V'}.$$

The following theorem is the main result of this Section. The result is proved in [27] in the Hilbert spaces setting and we have generalized it for Banach spaces.

Theorem C.7. Let V, $\tilde{V}$, Q and $\tilde{Q}$ be reflexive Banach spaces. Let $b\colon Q \times \tilde{V} \to \mathbb{R}$ be a bilinear form. Let $\tilde{Z} = \left\{ \tilde{z} \in \tilde{V} \middle| b(q, \tilde{z}) = 0 \text{ for all } q \in Q \right\}$ and define $\tilde{W} = \tilde{V}/\tilde{Z}$. Assume that there exists $M > 0$ such that

$$b(q, \tilde{v}) \leqslant M \|q\|_Q \|\tilde{v}\|_{\tilde{V}} \quad \text{for all } q \in Q \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q \in Q \setminus \{0\}} \frac{b(q, \tilde{w})}{\|q\|_Q} \geqslant \beta \|\tilde{w}\|_{\tilde{W}} \quad \text{for all } \tilde{w} \in \tilde{W}$$

and that if $q \in Q$ satisfies

$$b(q, \tilde{v}) = 0 \quad \text{for all } \tilde{v} \in \tilde{V}$$

then $q = 0$. Let $c \colon \tilde{Q} \times V \to \mathbb{R}$ be a bilinear form. Let $Z = \left\{ z \in V \middle| b(\tilde{q}, z) = 0 \text{ for all } \tilde{q} \in \tilde{Q} \right\}$ and define $W = V/Z$. Assume that there exists $N > 0$ such that

$$c(\tilde{q}, v) \leqslant N \|\tilde{q}\|_{\tilde{Q}} \|v\|_V \quad \text{for all } \tilde{q} \in \tilde{Q} \text{ and } v \in V.$$

Assume also that there exists $\gamma > 0$ such that

$$\sup_{\tilde{q} \in \tilde{Q} \setminus \{0\}} \frac{c(\tilde{q}, w)}{\|\tilde{q}\|_{\tilde{Q}}} \geqslant \gamma \|w\|_W \quad \text{for all } w \in W$$

and that if $\tilde{q} \in \tilde{Q}$ satisfies

$$c(\tilde{q}, v) = 0 \quad \text{for all } v \in V$$

then $\tilde{q} = 0$. Let $a \colon V \times \tilde{V} \to \mathbb{R}$ be a bilinear form. Assume that there exists a constant L such that

$$a(v, \tilde{v}) \leqslant L \|v\|_V \|\tilde{v}\|_{\tilde{V}} \quad \text{for all } v \in V \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\alpha > 0$ such that

$$\sup_{\tilde{v} \in \tilde{V} \setminus \{0\}} \frac{a(z, \tilde{v})}{\|\tilde{v}\|_{\tilde{V}}} \geqslant \alpha \|z\|_V \quad \text{for all } z \in Z$$

and that if $\tilde{z} \in \tilde{Z}$ satisfies

$$a(v, \tilde{z}) = 0 \quad \text{for all } v \in V$$

then $\tilde{z} = 0$. Then for all $f \in \tilde{V}'$ and $g \in \tilde{Q}'$ there exists a unique $(u, p) \in V \times Q$ such that

$$\begin{aligned} a(u, \tilde{v}) + b(p, \tilde{v}) &= \langle f, \tilde{v} \rangle \quad \text{for all } \tilde{v} \in \tilde{V} \\ c(\tilde{q}, u) &= \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}. \end{aligned}$$

Moreover

$$\begin{aligned} \|u\|_V &\leqslant \frac{1}{\gamma} \left(1 + \frac{L}{\alpha} \right) \|g\|_{\tilde{Q}'} + \frac{1}{\alpha} \|f\|_{\tilde{V}'} \\ \|p\|_Q &\leqslant \frac{1}{\beta} \left(1 + \frac{L}{\alpha} \right) \|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma} \left(1 + \frac{L}{\alpha} \right) \|g\|_{\tilde{Q}'}. \end{aligned}$$

Proof. By Theorem C.3 there exists a unique $w \in W$ such that

$$c(\tilde{q}, w) = \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}$$

and

$$\|w\|_W \leqslant \frac{1}{\gamma} \|g\|_{\tilde{Q}'}.$$

We identify w with one of its representative in V and we can choose it such that $\|w\|_V = \|w\|_W$. We notice that

$$a(w, \tilde{v}) \leqslant L \|w\|_V \|\tilde{v}\|_{\tilde{V}}.$$

Then $a(w,\cdot) \in \tilde{V}' \subset \tilde{Z}'$. Then $\langle f,\cdot\rangle - a(w,\cdot) \in \tilde{Z}'$. Then by Theorem C.3 the equation

$$a(z,\tilde{z}) = \langle f,\tilde{z}\rangle - a(w,\tilde{z}) \quad \text{for all } \tilde{z} \in \tilde{Z}$$

has a unique solution $z \in Z$ that satisfies

$$\|z\|_V \leqslant \frac{1}{\alpha}\left(\|f\|_{\tilde{V}'} + \frac{L}{\gamma}\|g\|_{\tilde{Q}'}\right).$$

Now define $u = w + z \in V$ and we notice that

$$a(u,\tilde{z}) = \langle f,\tilde{z}\rangle \quad \text{for all } \tilde{z} \in \tilde{Z} \tag{C.1.1}$$

and

$$\|u\|_V \leqslant \frac{1}{\gamma}\left(1 + \frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'} + \frac{1}{\alpha}\|f\|_{\tilde{V}'}.$$

Now we notice by (C.1.1) we have

$$\langle f,\tilde{z}\rangle - a(u,\tilde{z}) = \langle f,\tilde{z}\rangle - a(w,\tilde{z}) - a(z,\tilde{z}) = 0 \quad \text{for all } \tilde{z} \in \tilde{Z}.$$

Then by Remark C.6 we have $\langle f,\cdot\rangle - a(u,\cdot) \in \tilde{W}'$ and the equation

$$b(p,\tilde{v}) = \langle f,\tilde{v}\rangle - a(u,\tilde{v}) \quad \text{for all } v \in \tilde{V}$$

has a unique solution $p \in Q$ that satisfies

$$\|p\|_Q \leqslant \frac{1}{\beta}\left(\|f\|_{\tilde{V}'} + L\|u\|_V\right) \leqslant \frac{1}{\beta}\left(1 + \frac{L}{\alpha}\right)\|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma}\left(1 + \frac{L}{\alpha}\right)\|g\|_{\tilde{Q}'}.$$

This concludes the proof. □

C.2 Nonlinear functional analysis

C.2.1 Other

Lemma C.8. Let X and Y be Banach spaces. Let U an open subset of X and $F\colon U \to Y'$. Let $x \in U$. Assume that there exists $A \in \mathcal{L}(X,Y')$ and for all $v \in X$ there exists $\omega\colon \mathbb{R} \to [0,\infty)$ with $\omega(\epsilon) \to 0$ as $\epsilon \to 0$ such that

$$\left|\left\langle \frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av, y\right\rangle\right| \leqslant \omega(\epsilon)\|y\|_Y$$

for all $y \in Y$ and $\epsilon \in \mathbb{R}$ sufficiently small. Then F is Gateaux differentiable in x with Gateaux differential A.

Proof. Fix $v \in X$. Then by hypotheses there exists $\omega\colon \mathbb{R} \to [0,\infty)$ with $\omega(\epsilon) \to 0$ as $\epsilon \to 0$ such that

$$\left\|\frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av\right\|_{Y'} = \sup_{y\in Y\setminus\{0\}} \frac{\left\langle \frac{F(x+\epsilon v)-F(x)}{\epsilon} - Av, y\right\rangle}{\|y\|_Y} \leqslant \omega(\epsilon).$$

Then

$$\left\|\frac{F(x+\epsilon v) - F(x)}{\epsilon} - Av\right\|_{Y'} \to 0$$

as $\epsilon \to 0$. This is exactly the thesis. □

Lemma C.9. Let X and Y be two Banach spaces. Let A_n and A in $\mathcal{L}(X, Y')$ such that for all $x \in X$ and $y \in Y$ we have

$$|\langle A_n x - Ax, y\rangle| \leqslant \omega_n \|x\|_X \|y\|_Y$$

where $\omega_n \to 0$. Then $A_n \to A$ in $\mathcal{L}(X, Y')$.

Proof. We have

$$\|A_n - A\|_{\mathcal{L}(X,Y')} = \sup_{x \in X\setminus\{0\}} \frac{\|A_n x - Ax\|_{Y'}}{\|x\|_X} = \sup_{x\in X\setminus\{0\}} \sup_{y \in Y\setminus\{0\}} \frac{\langle A_n x - Ax, y\rangle}{\|x\|_X \|y\|_Y} \leqslant \omega_n \to 0.$$

This concludes the proof. □

C.2.2 Implicit function theorem

Theorem C.10. Let X, Y and E be Banach spaces. Let U and open set of $X \times Y$ and $(x_0, y_0) \in U$. Let $F \in C^1(U, E)$ such that $F(x_0, y_0) = 0$ and $D_x F(x_0, y_0) \in \mathrm{Iso}(X, E)$. Then we can find $r > 0$ and $s > 0$ with $B(x_0, r) \times B(y_0, s) \subset U$ such that there exist a unique $\chi \in C^1(B(y_0, s), B(x_0, r))$ satisfying

$$F(\chi(y), y) = 0 \quad \text{for all } y \in B(y_0, s).$$

Moreover if F is of class C^r with $r \geqslant 2$ also χ is of class C^r.

Proof. See [28]. □

C.2.3 Fixed point theorems

Theorem C.11 (Schauder fixed point theorem)**.** Let X be a Banach space and $C \subset X$ closed and convex. Let $F\colon C \to C$ continuous and with compact range. Then F admits a fixed point.

Proof. See [28]. □

C.2.4 Local existence theorem for Cauchy problems in Banach spaces

Theorem C.12. Let X be a Banach space and let U be an open subset of $X \times \mathbb{R}$ with $(x_0, 0) \in U$. Let $G \in C^1(U, X)$. Then there exists $T > 0$ such that the Cauchy problem

$$\begin{cases} \dot{x}(t) = G(x(t), t) & \text{for } t \in [0, T] \\ x(0) = x_0 \end{cases}$$

admits a unique solution $x \in C^2([0, T], X)$.

Proof. Since G is of class C^1 there there exists a neighborhood $V \subset U$ of $(x_0, 0)$ such that G and $D_x G$ are bounded in V. Let $M = \sup_V \|G\|_X < \infty$ and $L = \sup_V \|D_x G\|_{\mathcal{L}(X,X)}$. We notice that G is Lipschitz continuous with respect to x and uniformly in t on V with Lipschitz constant L. Let $B = B(x_0, r)$ for some $r > 0$ and $T > 0$ such that $B(x_0, r) \times (-T, T) \subset V$. We introduce the normed space $\mathcal{X} = C^0([0, T], B)$ endowed with the norm $\|x\|_{\mathcal{X}} = \sup_{t\in[0,T]} \|x(t)\|_X$. Since $\mathcal{X}$ is a closed subset of $C^0([0, T], B)$ it is a complete metric space. Now introduce the map

$$T\colon B \to C^0([0, T], B)$$

defined by
$$T(x)(t) = x_0 + \int_0^t G(x(s), s)ds.$$
We notice that $\|T(x) - x_0\|_X \leqslant TM$. Up to choose a smaller T we can assume that $TM < r$ so that actually $T\colon B \to B$. Moreover for any x_1 and x_2 in X we have
$$\|T(x_2) - T(x_1)\|_{\mathcal{X}} \leqslant \int_0^T \|G(x_2(s), s) - G(x_1(s), s)\|_X ds \leqslant TL\|x_2 - x_1\|_{\mathcal{X}}.$$
Up to choose a smaller T we can assume that $TL < 1$. In this case T is a strict contraction. By the Banach–Caccioppoli fixed point theorem there exists a unique $x \in B$ such that $Tx = x$. Since Tx is of class C^2 then also x is of class C^2. □

C.2.5 Browder–Minty theorem

The results of this section are taken from [32] and [33].

Let X be a Banach space and $T\colon X \to X'$ be a nonlinear operator and $f \in X'$. The Browder–Minty theorem allows to study the existence of solutions to the equation $T(x) = f$ when there is not any variational formulation. The key assumption is that T is coercive and monotone, a property that makes T similar in some sense to an elliptic operator.

Definition C.13. Let X be a Banach space and let $T\colon X \to X'$. We say that

(i) T is bounded if maps bounded sets of X into bounded sets of X',

(ii) T is coercive if
$$\frac{\langle T(x), x\rangle}{\|x\|} \to \infty \quad \text{as } \|x\| \to \infty$$
and that

(iii) T is monotone if
$$\langle T(x) - T(y), x - y\rangle \geqslant 0 \quad \text{for all } x \text{ and } y \text{ in } X.$$

We now prove a lemma that establish the equivalence between the equation $T(x) = f$ and a variational inequality under the assumption that T is monotone.

Lemma C.14. Let $T\colon X \to X'$ be continuous and $f \in X'$. Then if $x \in X$ is a solution to the variational inequality
$$\langle T(y) - f, y - x\rangle \geqslant 0 \quad \text{for all } y \in X$$
it is also a solution to te equation $T(x) = f$. Moreover if T is monotone then any solution to the equation $T(x) = f$ is also a solution to the variational inequality.

Proof. Let $y = x + tv$ with $t > 0$ and $v \in X$. Then if $x \in X$ is a solution to the variational inequality we have
$$\langle T(x + tv) - f, v\rangle \geqslant 0.$$
Letting $t \to 0$ we have $\langle T(x) - f, v\rangle \geqslant 0$. By arbitrariness of v we have $T(x) = f$. Conversely let $x \in X$ be such that $T(x) = f$. By monotonicity
$$\langle T(y) - f, y - x\rangle = \langle T(y) - T(x), y - x\rangle \geqslant 0.$$
Then x is a solution to the variational inequality. □

Now we study the equation $T(x) = f$ when the Banach space is finite dimensional. This results will be used to get the general result using the Galerkin method.

Proposition C.15. Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be continuous and coercive. Then for every $f \in \mathbb{R}^n$ the equation $T(x) = f$ has al least one solution.

Proof. Without loss of generality we cn assume that $f = 0$. Indeed otherwise define $S(x) = T(x) - f$ and notice that S is still continuous and coercive. Indeed

$$\frac{S(x)\cdot x}{|x|} = \frac{T(x)\cdot x}{|x|} - \frac{f\cdot x}{|x|} \geqslant \frac{T(x)\cdot x}{|x|} - |f|.$$

Now define $G\colon \mathbb{R}^n \to \mathbb{R}^n$ by $G(x) = x - T(x)$. We notice that

$$G(x)\cdot x = |x|^2 - T(x)\cdot x.$$

Since T is coercive there exists an $r > 0$ such that $|x|^2 - G(x)\cdot x = T(x)\cdot x > 0$ for all $x \in \mathbb{R} \setminus B(0, r)$. We define $\Pi\colon \mathbb{R}^n \to C$ with $C = \overline{B}(0, r)$ by

$$\Pi(x) = \begin{cases} x & \text{if } |x| \leqslant r \\ r\frac{x}{|x|} & \text{if } |x| > r. \end{cases}$$

We set $H(x) = \Pi(G(x))$ and we notice that $H\colon C \to C$ and it is continuous. By Brouwer fixed point theorem there exists $x \in C$ such that $H(x) = x$. If $x \in \mathring{C}$ then $x = H(x) = \Pi(G(x)) = G(x)$ so that $T(x) = 0$. If we prove that $x \notin \partial C$ we have concluded. By contradiction assume that $|x| = r$. Then $|G(x)| \geqslant r$ and

$$|x|^2 = x\cdot x = H(x)\cdot x = \frac{r}{|G(x)|}G(x)\cdot x < \frac{r}{|G(x)|}|x|^2 \leqslant |x|^2$$

where we used that $|x|^2 > G(x)\cdot x$ for $|x| \geqslant r$. This is the a contradiction. $\square$

We are now in position to prove the Browder–Minty theorem.

Theorem C.16. Let X be a reflexive and separable Banach space. Let $T\colon X \to X'$ be bounded, continuous, coercive and monotone. Then for every $f \in X'$ there exists a solution $x \in X$ to the equation $T(x) = f$.

Proof. As anticipated the proof relies on the Galerkin method.

Step 1. Let $\{y_n\} \subset X$ such that the linear combinations of y_n are dense in X. Let

$$X_n = \operatorname{span}\{y_m\}_{m=1}^n.$$

We define $f_n \in X_n'$ by $\langle f_n, x\rangle = \langle f, x\rangle$ for all $x \in X_n$. We also define $T_n\colon X_n \to X_n'$ by $\langle T_n(x), y\rangle = \langle T(x), y\rangle$ for all x and y in X_n.

Step 2. We prove that T_n are still bounded, continuous and coercive. For all $x \in X_n$ we have

$$\|T_n(x)\|_{X_n'} = \sup_{y\in X_n\setminus\{0\}} \frac{\langle T_n(x), y\rangle}{\|y\|_{X_n}} \leqslant \sup_{y\in X\setminus\{0\}} \frac{\langle T(xx), y\rangle}{\|y\|_X} = \|T(x)\|_X$$

so the boundedness of T_n follows form the boundedness of T. Moreover let $x_k \to x$ in X_n. Then

$$\|T_n(x_k) - T_n(x)\|_{X_n'} = \sup_{y\in X_n\setminus\{0\}} \frac{\langle T_n(x_k) - T_n(x), y\rangle}{\|y\|_{X_n}} \leqslant \sup_{y\in X\setminus\{0\}} \frac{\langle T(x_k) - T(x), y\rangle}{\|y\|_X} =$$
$$= \|T(x_k) - T(x)\|_{X'} \to 0$$

and this proves that T_n is continuous. Moreover for $x \in X_n$

$$\frac{\langle T_n(x), x\rangle}{\|x\|_{X_n}} = \frac{\langle T(x), x\rangle}{\|x\|_X}$$

so the coercivity of T_n follows immediately from the coercivity of T. By C.15 we have that for each n there exists $x_n \in X_n$ that solves $T_n(x_n) = f_n$.

Step 3. We have the estimate

$$\frac{\langle T(x_n), x_n\rangle}{\|x_n\|_X} \leqslant \frac{\langle T_n(x_n), x_n\rangle}{\|x_n\|_X} = \frac{\langle f_n, x_n\rangle}{\|x_n\|_X} = \frac{\langle f, x_n\rangle}{\|x_n\|_X} \leqslant \|f\|.$$

Since T is coercive then x_n are bounded in X and since T is bounded then $T(x_n)$ are bounded in X'. Then up to a subsequence

$$x_n \rightharpoonup x \quad \text{in } X$$
$$T(x_n) \overset{*}{\rightharpoonup} g \quad \text{in } X'$$

for some $x \in X$ and $g \in X'$.

Step 4. We first prove that $g = f$. To this aim we fix any y_m and we notice that

$$\langle g - f, y_m\rangle = \lim_{n\to\infty} \langle T(x_n) - f, y_m\rangle = 0.$$

Then by density we have indeed $g = f$. Next we have

$$\langle T(x_n), x_n\rangle = \langle T_n(x_n), x_n\rangle = \langle g_n, x_n\rangle = \langle g, x_n\rangle \to \langle g, x\rangle.$$

Step 5. Eventually using the monotonicity of T we have

$$0 \geqslant \langle T(y) - T(x_n), y - x_n\rangle \to \langle T(y) - g, y - x\rangle = \langle T(y) - f, y - x\rangle.$$

Applying Lemma C.14 we get the thesis. □

Actually to obtain the same existence result the monotonicity assumption can be weakened.

Definition C.17. Let X be a Banach space and let $T\colon X \to X'$. We say that T is pseudomonotone if it is bounded and every time

$$x_k \rightharpoonup x \text{ in } X \quad \text{and} \quad \limsup_{k\to\infty} \langle T(x_k), x_k - x\rangle \leqslant 0$$

then

$$\liminf_{k\to\infty} \langle T(x_k), x_k - y\rangle \geqslant \langle T(x), x - y\rangle$$

for all $y \in X$.

Theorem C.18. Let X be a separable and reflexive Banach space. Let $T\colon X \to X'$ be continuous, coercive and pseudomonotone. Then for every $f \in X'$ there exists $x \in X$ such that $T(x) = f$.

Proof. The proof is exactly the same as the one of Theorem C.16 except to Step 5. We know that $x_k \rightharpoonup x$ in X, that $T(x_k) \overset{*}{\rightharpoonup} f$ in X' and $\langle T(x_k), x_k\rangle \to \langle f, x\rangle$. Then in particular

$$\langle T(x_k), x_k - x\rangle = \langle T(x_k), x_k\rangle - \langle T(x_k), x\rangle \to \langle f, x\rangle - \langle f, x\rangle = 0.$$

Since T is pseudomonotone then

$$\liminf_{k\to\infty}\langle T(x_k), x_k - y\rangle \geqslant \langle T(x), x - y\rangle$$

for all $y \in X$. But

$$\liminf_{k\to\infty}\langle T(x_k), x_k - y\rangle = \langle f, x - y\rangle.$$

Then

$$\langle f, x - y\rangle \geqslant \langle T(x), x - y\rangle$$

for all $y \in X$. By arbitrariness of y we have $T(x) = f$. □

The following proposition establish the relationship between monotone and pseudomonotone operators.

Proposition C.19. Let X be a Banach space. Let $T \colon X \to X'$ be continuous, bounded and monotone. Then T is pseudomonotone.

Proof. Let $x_k \rightharpoonup x$ with

$$\limsup_{k\to\infty}\langle T(x_k), x_k - x\rangle \leqslant 0.$$

Since T is monotone then

$$\langle T(x_k), x_k - x\rangle \geqslant \langle T(x), x_k - x\rangle \to 0.$$

This implies also

$$\liminf_{k\to\infty}\langle T(x_k), x_k - x\rangle \geqslant 0.$$

Then

$$\lim_{k\to\infty}\langle T(x_k), x_k - x\rangle = 0.$$

Now let $y \in X$ and set $y_\epsilon = (1-\epsilon)x + \epsilon y$ for $\epsilon > 0$. By monotonicity

$$0 \leqslant \langle T(x_k) - T(y_\epsilon), x_k - y_\epsilon\rangle = \langle T(x_k) - T(y_\epsilon), \epsilon(x - y) + x_k - x\rangle.$$

Rearranging the terms we have

$$\epsilon\langle T(x_k), x - y\rangle \geqslant \langle T(y_\epsilon), x_k - x\rangle - \langle T(x_k), x_k - x\rangle + \epsilon\langle T(y_\epsilon), x - y\rangle.$$

Passing to le limit $k \to \infty$ with $\epsilon > 0$ fixed

$$\epsilon \liminf_{k\to\infty}\langle T(x_k), x - y\rangle \geqslant \epsilon\langle T(y_\epsilon), x - y\rangle.$$

Dividing by ϵ and passing to the limit $\epsilon \to 0^+$ we obtain using the continuity of T that

$$\liminf_{k\to\infty}\langle T(x_k), x - y\rangle \geqslant \langle T(x), x - y\rangle.$$

Eventually

$$\liminf_{k\to\infty}\langle T(x_k), x_k - y\rangle = \lim_{k\to\infty}\langle T(x_k), x_k - x\rangle + \liminf_{k\to\infty}\langle T(x_k), x - y\rangle \geqslant 0.$$

This proves that T is pseudomonotone. □

The following result is useful.

Proposition C.20. Let X be a Banach space. Let T_1 and T_2 from X into X' be pseudomonotone operators. Then also $T_1 + T_2$ is pseudomonotone.

Proof. Let $x_k \rightharpoonup x$ with

$$\limsup_{k\to\infty} \langle T_1(x_k) + T_2(x_k), x_k - x\rangle \leqslant 0.$$

We claim that

$$\limsup_{k\to\infty} \langle T_i(x_k), x_k - x\rangle \leqslant 0 \tag{C.2.1}$$

for both $i = 1$ and $i = 2$. By contradiction

$$\limsup_{k\to\infty} \langle T_2(x_k), x_k - x\rangle = \epsilon > 0.$$

Up to taking a subsequence we can assume that

$$\lim_{k\to\infty} \langle T_2(x_k), x_k - x\rangle = \epsilon > 0.$$

Then

$$\limsup_{k\to\infty} \langle T_1(x_k), x_k - x\rangle \leqslant -\epsilon < 0. \tag{C.2.2}$$

Then since T_1 is pseudomonotone

$$\liminf_{k\to\infty} \langle T_1(x_k), x_k - x\rangle \geqslant 0$$

but this contradicts (C.2.2). Then (C.2.1) holds. Since T_i is pseudomonotone then for all $y \in X$ we have

$$\liminf_{k\to\infty} \langle T_i(x_k), x_k - y\rangle \geqslant \langle T_i(x), x - y\rangle.$$

It follows that

$$\liminf_{k\to\infty} \langle T_1(x_k) + T_2(x_k), x_k - y\rangle \geqslant \langle T_1(x) + T_2(x), x - y\rangle.$$

This means that $T_1 + T_2$ is pseudomonotone. □

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